Transcendence of Algebraic Arc Lengths and Inverse Periods

Division Dichotomies, Beta Quantiles, and Arithmetic Transfers

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Abstract

We develop arithmetic consequences of algebraic-integral transcendence, consolidating the research discussion subsequent to *Explicit Transcendence Criteria for Beta-Type Integrals*. For noncircular polynomial lemniscates, the square root of the complexified Euclidean metric becomes a holomorphic differential on a finite cover. This yields transcendence and logarithmic separation for closed lengths and arcs with algebraic endpoints. We classify rational division parameters for all positive rational sinusoidal-spiral exponents, equivalently all rational quantiles of the beta distributions with parameters $(\alpha, 1/2)$, $\alpha \in \mathbb{Q}_{>0}$. Torsion-packet results strengthen selected cases to all algebraic division ratios. For a general positive regular differential, a minimal abelian quotient separates elliptic division from a finite set of algebraic exceptions; a division-hull argument treats algebraic, rather than merely rational, ratios. Explicit Cassini and hyperelliptic constructions give infinite families linearly independent over $\overline{\mathbb{Q}}$. Further transfers concern quadrature, orthogonal polynomials, probability, differential equations, and spectral models. Established transcendence and arithmetic-geometric theorems are identified as inputs. Linear independence is distinguished from algebraic independence, and unresolved constant problems remain unresolved.

Keywords. Abelian integrals; generalized Jacobians; lemniscates; beta quantiles; transcendence; analytic subgroup theorem; torsion points; quadrature.

Scope, provenance, and conventions

This sequel covers the discussion after the manuscript `Transcendence_Beta_Gamma_Unified.pdf` (11 October 2026). Necessary inputs are restated, and overlapping applications are grouped under common theorems. The numbered observations are not a count of independent research breakthroughs.

This is a research manuscript; neither external peer review nor historical priority is claimed. The analytic subgroup theorem, logarithmic and E-function theorems, torsion classifications, and arithmetic finiteness theorems are established inputs, identified in the proofs. Neither Euler's constant nor an unresolved individual Gamma value such as $\Gamma(1/5)$ is proved transcendental here.

We fix $\overline{\mathbb{Q}} \subset \mathbb{C}$. Real algebraic data lie in $\overline{\mathbb{Q}} \cap \mathbb{R}$, and positive branches are used on real integration intervals. Curves mean smooth projective models unless specified; "regular" means globally holomorphic. An oval is a real connected component, not necessarily an elliptic curve. Traversals at singular plane points are interpreted on the normalization. Linear independence means independence over $\overline{\mathbb{Q}}$, and an infinite family is independent when every finite subfamily is. This differs from algebraic independence. Algebraicity of a point means algebraicity of all its coordinates; coordinate-wise transcendence requires its own proof.

Contents

Scope, provenance, and conventions 1

I Algebraic lengths and their arithmetic 3

1 Polynomial lemniscates: holomorphic length differentials 3

2 Closed-period transfers to other geometric quantities 7

3 An infinite linearly independent family of Cassini perimeters 9

II Division and inverse periods 11

4 Arithmetic division of real Abelian integrals 11

5 A complete rational-quantile classification for the beta family 17

6 Algebraic levels and torsion packets for generalized lemniscatic integrals 25

III Explicit independent Abelian integrals 29

7 Independent hyperelliptic integrals 29

8 Combining the two infinite families 36

IV Transfers to other mathematical structures 36

9 Arithmetic transfer principles and applications 36

A Register of the final division and independence results 49

B Boundaries of the conclusions and reproducibility 49

References 50

Part I
Algebraic lengths and their arithmetic

1 Polynomial lemniscates: holomorphic length differentials

The baseline argument identifies the length of a polynomial lemniscate with a nonzero closed period. The additional observation in this section is stronger: except for circles, the length differential can be made *holomorphic on a complete algebraic curve*. This removes the closed-path restriction, separates the resulting lengths from logarithms, and permits an explicit infinite linear-independence construction. Tamiozzo's Remark 8.16 asks whether all polynomial lemniscates over $\overline{\mathbb{Q}}$ have transcendental length; the examples proved there include $|z^k - a^k| = 1$ [1]. We give the geometric construction explicitly, including critical levels. The transcendence inputs throughout are established forms of the analytic subgroup theorem, not new versions of that theorem.

Write $\mathbb{R}_{\mathrm{alg}} = \overline{\mathbb{Q}} \cap \mathbb{R}$. Let

$$P \in \overline{\mathbb{Q}}[z] \setminus \overline{\mathbb{Q}}, \qquad \rho \in \mathbb{R}_{\mathrm{alg}}, \quad \rho > 0, \qquad \Lambda(P, \rho) = \{z \in \mathbb{C} : |P(z)| = \rho\}. \tag{1.1}$$

Length always means ordinary Euclidean geometric length, with overlapping arcs counted once. A finite arc family below means finitely many arcs whose interiors are disjoint up to finitely many points. An algebraic endpoint has both Cartesian coordinates in $\mathbb{R}_{\mathrm{alg}}$.

1.1 Normalization and the positive square-root lift

After replacing P by P/ρ, assume that $\rho = 1$. If d is the greatest common divisor of the multiplicities of the distinct roots, choose $Q \in \overline{\mathbb{Q}}[z]$ with $P = Q^d$. Then $\Lambda(P, 1) = \Lambda(Q, 1)$ and the root multiplicities $m_1, \dots, m_r$ of Q have greatest common divisor one. Put $n = \deg Q$ and let $Q^{\#}(w)$ denote the polynomial obtained by conjugating the coefficients of Q.

Lemma 1.1 (The underlying curve)**.** *The affine complex curve*

$$Q(z)Q^{\#}(w) = 1 \tag{1.2}$$

is geometrically irreducible. Its smooth projective normalization X has a real structure under $z = x + iy$, $w = x - iy$. Its real locus is a finite union of circles, all lying over finite points of the affine plane. On each such circle, the map to the plane is an immersion, including over critical points of Q on the lemniscate.

Proof. Work over $\overline{\mathbb{Q}}(z)$ and consider the degree-n polynomial $Q^{\#}(w) - 1/Q(z)$, scaled to be monic. At a zero z_j of multiplicity m_j, every root ξ has valuation $-m_j/n$: its valuation must be negative, and the leading term of $Q^{\#}(\xi)$ is then uniquely of least valuation. If a monic factor has degree e, its nonzero constant term therefore has valuation $-em_j/n$. This is an integer, so n divides em_j for every j. Since $\gcd_j m_j = 1$, n divides e. A proper factor is impossible.

The leading homogeneous part of the real plane equation is $|a_n|^2(x^2 + y^2)^n$, so its projective closure has no real point at infinity. Consequently the real normalization is compact and its components are circles. To examine a point z_0 on the level set, take a local holomorphic logarithm of $Q(z)/Q(z_0)$. It vanishes to some order $m \geq 1$ and, after a biholomorphic change of local coordinate, equals u^m. The real level set is thus $\mathrm{Re}(u^m) = 0$, a union of m distinct real lines. The normalization separates these branches, and the inverse coordinate change has nonzero derivative on each. Hence the normalized map is an immersion. This also shows that every affine real point of the lemniscate is covered by real branches of the normalization. □

On X consider the rational quadratic differential

$$q = dz\,dw = dx^2 + dy^2. \tag{1.3}$$

Choose a nonzero rational differential τ on X, write $q = f\tau^2$, and normalize the quadratic extension obtained by adjoining $\sqrt{f}$. If f is already a square, use X itself and choose a square root. We obtain a connected smooth projective curve Y over $\overline{\mathbb{Q}}$, a map $\pi : Y \to X$ of degree at most two, and a rational differential ω satisfying

$$\omega^2 = \pi^* q. \tag{1.4}$$

All choices here are algebraic. On the real normalization circles, q is strictly positive on nonzero real tangent vectors. Thus the cover is unramified there, and, after orienting a circle, exactly one local lift has ω positive on its tangent. These local lifts join to a closed lift: the positive choice is unique when the initial point and tangent return. In the degree-one case one chooses the orientation giving a positive integral. In either case its integral is its Euclidean length.

Proposition 1.2 (Closed lengths)**.** *The length of every geometric connected component of $\Lambda(P, \rho)$, and its total length, is a nonzero period of a rational differential on an algebraic curve over $\overline{\mathbb{Q}}$. Consequently each is transcendental.*

Proof. The normalization is one-to-one away from finitely many singular points. For a geometric connected component, take all the normalization circles lying above it, lift them as just described, and add their cycles. Their integrals give the geometric length, since the exceptional finite set has length zero. This integral is positive. Wüstholz's theorem that a nonzero closed period of an algebraic rational differential is transcendental applies; see [1, Theorem 8.14] and [2]. Repeated factors were removed without changing the geometric set, so no multiplicity is counted. □

1.2 The additional holomorphy statement

Lemma 1.3 (Holomorphy away from the circle case)**.** *If P has at least two distinct roots, the differential ω in (1.4) is holomorphic everywhere on Y.*

Proof. At finite points of X, both z and w are regular, so $q = dz\,dw$ has no pole. To inspect infinity, use the closure of (1.2) in $\mathbb{P}^1 \times \mathbb{P}^1$. If z has a pole at a point of X, the defining equation forces w to tend to a finite root β of $Q^{\#}$. Write

$$\operatorname{ord}(z) = -a, \qquad \operatorname{ord}(w - \beta) = b, \qquad a, b > 0,$$

and let m be the multiplicity of β. Comparing valuations gives $na = mb$. Because Q has at least two distinct roots, $m < n$, whence $b > a$. In characteristic zero,

$$\operatorname{ord}(q) = (-a - 1) + (b - 1) = b - a - 2 \geq -1. \tag{1.5}$$

The case where w has a pole is identical. These exhaust the points at infinity; the two coordinates cannot both have poles.

If a quadratic differential has local order $k \geq -1$, an even k gives an unramified square root of order $k/2 \geq 0$. For odd k, the square-root cover has local parameter $t = s^2$, and the pulled-back quadratic differential has order $2k + 2$. Its square root has order $k + 1 \geq 0$. Thus even a possible simple pole of q produces a regular ω. □

The calculation at infinity is essential. The fact that a length is a closed period alone would not justify replacing its cycle by an open arc. Lemma 1.3 supplies precisely the stronger input needed for that replacement.

1.3 Separation of Abelian integrals from logarithms

Define the $\overline{\mathbb{Q}}$-vector space

$$\mathcal{L} = \overline{\mathbb{Q}} + \operatorname{span}_{\overline{\mathbb{Q}}}\{\lambda \in \mathbb{C} : \exp(\lambda) \in \overline{\mathbb{Q}}^{\times}\}. \tag{1.6}$$

All branches of logarithms are allowed; in particular $2\pi i \in \mathcal{L}$. A first-kind algebraic integral means an integral of a holomorphic differential defined over $\overline{\mathbb{Q}}$ on a smooth projective curve over $\overline{\mathbb{Q}}$, along a path with algebraic endpoints. Closed periods are included. Finite sums can equivalently be treated on a product of Jacobians.

Lemma 1.4 (Abelian–logarithmic separation)**.** *Let S be a $\overline{\mathbb{Q}}$-linear combination of first-kind algebraic integrals. If $S \in \mathcal{L}$, then $S = 0$. Thus every nonzero such S is transcendental and S/π is transcendental.*

Proof. On a product A of the relevant Jacobians, write $S = \eta(v)$, where η is an invariant algebraic differential and $\exp_A(v) \in A(\overline{\mathbb{Q}})$. This uses the algebraicity of the endpoint divisors. Suppose

$$\eta(v) + c_0 + \sum_{j=1}^{r} c_j \lambda_j = 0, \qquad c_j \in \overline{\mathbb{Q}}, \quad e^{\lambda_j} \in \overline{\mathbb{Q}}^{\times}.$$

Apply the refined analytic subgroup theorem [2, Theorem 6.2] to

$$G = A \times \mathbb{G}_a \times \mathbb{G}_m^r, \qquad u = (v, 1, \lambda_1, \dots, \lambda_r).$$

Its exponential is algebraic. The theorem gives a connected algebraic subgroup $H \leq G$ with $u \in \operatorname{Lie}(H)_{\mathbb{C}}$ and with the displayed algebraic linear form vanishing on $\operatorname{Lie}(H)$.

A connected subgroup of a product of an abelian variety and a connected affine commutative algebraic group splits between these two types. For example, the algebraic-group Goursat argument reduces a possible mixing to a common connected quotient, which would be both proper and affine and hence trivial. Accordingly $H = B \times H_{\mathrm{aff}}$ for an abelian subvariety $B \leq A$. Restricting the linear form to $\operatorname{Lie} B$ shows that η vanishes there. Since $v \in \operatorname{Lie}(B)$, $S = 0$. The last conclusions follow because both $\overline{\mathbb{Q}}$ and $\overline{\mathbb{Q}}\pi$ lie in $\mathcal{L}$. □

Theorem 1.5 (Algebraic-endpoint arcs and finite arc families)**.** *Suppose that P has at least two distinct roots. The length $L > 0$ of any finite arc family on $\Lambda(P, \rho)$ with algebraic endpoints is a finite sum of first-kind algebraic integrals. Consequently*

$$L \notin \mathcal{L}, \qquad L \text{ is transcendental}, \qquad L/\pi \text{ is transcendental}. \tag{1.7}$$

The same conclusions hold for each full geometric connected component and for the total length.

Proof. Split an arc, when necessary, at its finitely many critical crossings. Those crossing points are algebraic: their complex coordinate is a zero of P'. Normalize each resulting branch and choose the lift for which ω integrates positively. Every lift of an algebraic endpoint under these algebraic finite maps is algebraic. The resulting integrals are of the first kind by Lemma 1.3, and their sum is L. For complete components, the earlier closed positive lifts give the same conclusion directly. Positivity and Lemma 1.4 finish the proof. □

Corollary 1.6 (Individual lobes and boundary circuits)**.** *The conclusions of Theorem 1.5 apply to each positive-length lobe or finite boundary circuit of a polynomial lemniscate whose vertices are critical crossings. In particular, each bounded complementary region has transcendental boundary length.*

Proof. At a crossing, a planar circuit may switch normalized branches; it need not lift to a closed cycle. Instead, cut at its algebraic vertices and apply the finite-arc assertion. Boundary arcs are counted once when computing geometric boundary length. A boundary without vertices is a complete smooth component and was already covered. □

For a circle, positive arcs with algebraic endpoints are also transcendental in length, although (1.7) need not hold in full. Indeed the radius is algebraic and an arc length is its product with an angle θ such that $e^{i\theta}$ is algebraic. A nonzero algebraic linear combination of logarithms of algebraic numbers cannot be algebraic, by the additive–multiplicative case of the same subgroup theorem. Thus the transcendence assertion for algebraic endpoint arcs holds for *every* polynomial in (1.1).

Theorem 1.7 (Exactly when a perimeter is an algebraic multiple of π)**.** *Let L be the total length or the length of any geometric connected component of $\Lambda(P,\rho)$. Then*

$$L/\pi \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad P(z) = a(z-b)^n \text{ for some } a \in \overline{\mathbb{Q}}^{\times},\ b \in \overline{\mathbb{Q}},\ n \geq 1.$$

Equivalently, this is precisely the circle case.

Proof. If $P = a(z-b)^n$, the level is the circle of radius $R = (\rho/|a|)^{1/n} \in \mathbb{R}_{\mathrm{alg}}$, so $L = 2\pi R$. Otherwise P has at least two distinct roots and Theorem 1.5 excludes algebraicity of L/π.

For completeness, the geometric description of the exception is also elementary. If a circle $z = b + R\zeta$, $|\zeta| = 1$, is contained in the level set, put $F(\zeta) = P(b + R\zeta)$ and $N = \deg F$. Define $F^*(\zeta) = \zeta^N \overline{F(1/\bar{\zeta})}$. The circle identity implies the polynomial identity $F(\zeta)F^*(\zeta) = \rho^2\zeta^N$. Hence F has no nonzero root, so $F(\zeta) = c\zeta^N$. Thus P has the asserted single-root form. This argument also excludes a circle component in a noncircle case. □

Corollary 1.8 (Exponentials and trigonometric values)**.** *Let $L > 0$ be any length covered by Theorem 1.5, and let $c \in \overline{\mathbb{Q}}^{\times}$. Then*

$$e^{cL}, \qquad \sin(cL), \qquad \cos(cL), \qquad \tan(cL)$$

are transcendental; the last expression is defined.

Proof. If e^{cL} were algebraic, cL would belong to the logarithmic space (1.6), contrary to Lemma 1.4. An algebraic sine or cosine would make e^{icL} a root of a quadratic polynomial over $\overline{\mathbb{Q}}$. Thus both are transcendental and in particular $\cos(cL) \neq 0$. Finally an algebraic tangent would make $e^{2icL} = (1 + i\tan(cL))/(1 - i\tan(cL))$ algebraic. The denominator cannot vanish for a finite exponential representation. Each alternative is impossible. □

Theorem 1.9 (Elementary integration occurs exactly for circles)**.** *The algebraic length differential of a polynomial lemniscate admits an elementary antiderivative, in the differential-algebraic sense, if and only if the lemniscate is a circle. Elementary extensions and algebraic changes of variable are allowed; a numerical special value of an integral is not what is being classified here.*

Proof. For a circle of algebraic radius R the length differential is, up to orientation, $-iR\,d\zeta/\zeta$, and its primitive is logarithmic. Otherwise Lemma 1.3 gives a nonzero holomorphic differential on a complete smooth curve. Such a differential has no elementary primitive. Indeed Liouville's theorem [5] would give, on a finite algebraic cover if necessary, an identity

$$\omega = dF + \sum_{j=1}^{r} c_j \frac{dG_j}{G_j}, \qquad c_j \in \mathbb{C},$$

with F, G_j meromorphic. A pole of F gives a pole of order at least two in dF, whereas logarithmic derivatives have only simple poles. Thus F is constant. Vanishing of the remaining residues means $\sum_j c_j \operatorname{div}(G_j) = 0$. This system has integer coefficients, so its complex kernel is spanned by rational relations. Clearing denominators in each rational relation makes the corresponding product of powers of the G_j constant, so its logarithmic derivative is zero. Consequently $\omega = 0$, a contradiction. Pullback to a finite cover preserves holomorphy and nonvanishing. □

Proposition 1.10 (Linear combinations of lengths)**.** *For finitely many polynomial lemniscate lengths, including the algebraic-endpoint arc lengths above, every $\overline{\mathbb{Q}}$-linear combination is either zero or transcendental. For lengths drawn only from polynomials with at least two distinct roots, a nonzero combination lies outside $\mathcal{L}$ and its quotient by π is transcendental.*

Proof. Separate the combination into first-kind integrals from noncircle lemniscates and logarithmic terms from circles. If the total were algebraic, Lemma 1.4 would force the first-kind part to vanish. The remaining logarithmic linear form is either zero or transcendental, so an algebraic total must be zero. The second assertion is exactly Lemma 1.4 on a product of Jacobians. □

These statements do not assert algebraic independence of lengths, or even linear independence of an arbitrary chosen collection. Isometries, algebraic dilations and symmetries create genuine dependencies. The next construction provides an explicit collection for which independence can be proved.

2 Closed-period transfers to other geometric quantities

The following extensions use rational differentials and closed cycles; they do not require the extra holomorphy for noncircle polynomial lemniscates. A finite algebraic linear combination of closed rational-differential periods, even on different curves, is either zero or transcendental. To justify this without assuming that sums of transcendental numbers are transcendental, puncture each curve at the poles, take their product, and use the closed algebraic 1-form $\sum_j c_j \operatorname{pr}_j^* \eta_j$ there. Its period on the corresponding homology class is the required sum. Wüstholz's theorem for smooth quasi-projective varieties applies [6, Theorem 3.1].

Theorem 2.1 (Rational lemniscates)**.** *Let $R \in \overline{\mathbb{Q}}(z)$ be nonconstant and $\rho > 0$ algebraic. Every bounded geometric connected component of $\{z \in \mathbb{C} : |R(z)| = \rho\}$ has transcendental Euclidean length. On the Riemann sphere, every connected component and the whole level set have transcendental length for the unit-sphere metric*

$$ds_{\mathrm{sph}} = \frac{2|dz|}{1 + |z|^2}.$$

The spherical level includes ∞ when $R(\infty)$ is finite and has modulus ρ.

Proof. Normalize the reduced algebraic correspondence $R(z)R^{\#}(w) = \rho^2$, component by component, taking its closure from the locus where both rational functions are defined and finite. Any irreducible component supporting a real arc is fixed by the real structure. At each level point, a local logarithm of $R/R(z_0)$ reduces the real equation to $\operatorname{Re}(u^m) = 0$. The normalized branches are therefore immersed, including at critical crossings and, in a spherical local coordinate, at infinity.

For a bounded Euclidean component the normalization circles remain in the finite plane; use $q = dz\,dw$. For spherical length use

$$q_{\mathrm{sph}} = \frac{4\,dz\,dw}{(1 + zw)^2}.$$

This is positive and nonsingular along the real locus. At infinity, $u = 1/z$, $v = 1/w$ give the identical expression $4\,du\,dv/(1+uv)^2$. Each positive square-root lift is closed, giving a rational-differential period. Sums over different normalization components are covered by the product argument above. Positivity proves transcendence. No finite Euclidean length is asserted for an unbounded component. □

For the next two results, an algebraic immersed closed curve means a real normalization circle γ on a smooth projective curve over $\mathbb{R}_{\mathrm{alg}}$, with rational coordinate functions x, y regular along γ, such that (dx, dy) never vanishes there. Integrals count one traversal. For an embedding, or a map generically one-to-one onto its image, this is geometric length without repeated arcs.

Theorem 2.2 (Weighted length and bending energies)**.** *For such a curve, let $W \in \mathbb{R}_{\mathrm{alg}}[x, y]$ be nonnegative on the image and not identically zero there. Then*

$$\int_\gamma W\,ds \quad \text{and} \quad \int_\gamma \kappa^{2m}\,ds \quad (m \geq 1)$$

are transcendental, where κ is Euclidean curvature.

Proof. Use the positive closed lift with algebraic differential $\omega = ds$. The weighted length is the positive period of the rational differential $W\omega$. For curvature, the derivation D defined by $df = (Df)\omega$ is algebraic on the square-root curve and is arc-length differentiation on the real lift. Hence

$$\kappa^2 = (D^2x)^2 + (D^2y)^2$$

is rational and regular along that lift. The bending energy is the period of $(\kappa^2)^m\omega$. It is positive, because identically zero curvature would make the unit tangent constant, which is incompatible with a nonconstant closed immersed curve. The closed-period theorem proves both assertions. □

Theorem 2.3 (Anisotropic length derivatives)**.** *For an algebraic immersed closed curve define*

$$A(t) = \int_\gamma \sqrt{t\,dx^2 + dy^2}, \qquad t > 0.$$

If x is nonconstant, then for every positive algebraic t and every integer $n \geq 0$, $A^{(n)}(t)$ is nonzero and transcendental. In this planar closed-immersion setting, nonconstancy of x is automatic.

Proof. Positivity of the metric on the compact immersed curve permits differentiation under the integral to every order. Fix an algebraic $t > 0$ and its positive closed square-root differential $\omega_t^2 = t\,dx^2 + dy^2$. For $n \geq 1$,

$$A^{(n)}(t) = c_n \int_{\widetilde{\gamma}} \left(\frac{dx}{\omega_t}\right)^{2n} \omega_t, \qquad c_n = \prod_{j=0}^{n-1} \left(\frac{1}{2} - j\right) \in \mathbb{Q}^\times. \tag{2.1}$$

The integrand is rational over $\overline{\mathbb{Q}}$ and pole-free on the cycle. The integral before multiplication by c_n is strictly positive because dx is not identically zero. It is consequently a nonzero period, as is the length for $n = 0$. If x were constant, the periodic real function y would have a critical point, contradicting immersion. Without that immersion assumption the coordinate hypothesis must be retained: otherwise all positive-order derivatives can vanish. □

Theorem 2.4 (Areas bounded by smooth algebraic ovals)**.** *Let $\Omega \subset \mathbb{R}^2$ be a nonempty bounded domain whose boundary is a finite disjoint union of smooth embedded algebraic ovals over $\mathbb{R}_{\mathrm{alg}}$. If $W \in \mathbb{R}_{\mathrm{alg}}[x, y]$ is nonnegative on Ω and not identically zero there, then*

$$\iint_\Omega W(x, y)\,dx\,dy$$

is transcendental. This includes the ordinary area.

Proof. The polynomial $F(x,y) = \int_0^x W(s,y)\,ds$ has algebraic coefficients. Green's theorem with the induced boundary orientation gives

$$\iint_\Omega W\,dx\,dy = \int_{\partial\Omega} F\,dy.$$

Each oval is a closed cycle on its algebraic normalization; $F\,dy$ is rational and has no pole on that cycle. The sum is strictly positive, so the product form of the closed-period theorem applies. Inner boundaries carry the opposite orientation. Positivity comes from the area integral, not from each separate boundary integral. □

The smooth-boundary hypothesis is substantive. The critical filled lemniscate $\Omega_* = \{z : |z^2 - 1| \le 1\}$ has area 2. Its polar equation gives $0 \le r^2 \le 2\cos(2\theta)$ where the cosine is nonnegative, and hence

$$\operatorname{area}(\Omega_*) = \int_{\{0\le\theta\le 2\pi:\cos(2\theta)\ge 0\}} \cos(2\theta)\,d\theta = 2.$$

The positive boundary orientation switches branches at the node and does not produce the required closed cycle on the normalization. Thus we do not assert transcendental area for every singular algebraic region.

3 An infinite linearly independent family of Cassini perimeters

For each odd prime p, let

$$C_p = \{z \in \mathbb{C} : |z^2 - p^{-2}| = 1\}, \qquad \ell_p = \operatorname{length}(C_p), \qquad m_p = \frac{1 - \sqrt{1 - p^{-4}}}{2}. \tag{3.1}$$

Each C_p is a smooth connected Cassini oval. We use the parameter, rather than the square-root modulus, convention

$$K(m) = \int_0^1 \frac{dt}{\sqrt{(1-t^2)(1-mt^2)}}.$$

The classical reduction of the Cassini perimeter, in the form proved in [1, Equation (7.7.4)], gives

$$\ell_p = 4K(m_p). \tag{3.2}$$

Thus ℓ_p is a nonzero complete period of the algebraic differential dx/y on the elliptic curve

$$E_p : \quad y^2 = x(1-x)(1-m_p x). \tag{3.3}$$

Indeed $\int_0^1 dx/y = 2K(m_p)$, and traversing this interval on one sheet and returning on the other doubles the integral.

Lemma 3.1 (Pairwise nonisogeny)**.** *The elliptic curves E_p, as p ranges over the odd primes, are pairwise nonisogenous over $\overline{\mathbb{Q}}$.*

Proof. The standard Legendre invariant is

$$j(m) = 256\frac{(1 - m + m^2)^3}{m^2(1-m)^2}.$$

Since $m_p(1 - m_p) = 1/(4p^4)$, direct substitution yields

$$j(E_p) = 64\frac{(4p^4 - 1)^3}{p^4} \in \mathbb{Q}. \tag{3.4}$$

At the rational prime p this has valuation -4. For a different odd prime q, the rational number $j(E_q)$ has nonnegative valuation at p.

An elliptic curve over a number field has potentially good reduction at a finite place if and only if its j-invariant is integral there; see [3, Chapter VII, Proposition 5.5]. Potentially good reduction is invariant under isogeny. One way to see the latter assertion is to pass to an extension with good reduction and use the Néron–Ogg–Shafarevich criterion: an isogeny identifies the rational ℓ-adic Tate modules for an auxiliary prime ℓ different from the residue characteristic [4].

If E_p and E_q were isogenous over $\overline{\mathbb{Q}}$, the isogeny and both curves would be defined over a common number field. At any place over p, E_q would have potentially good reduction and E_p would not. This is impossible. The argument allows arbitrary degrees of isogeny and all finite extensions of the field of definition. □

Theorem 3.2 (Cassini perimeter independence)**.** *The collection*

$$\{1, \pi\} \ \cup \ \{\ell_p : p \text{ an odd prime}\} \tag{3.5}$$

is linearly independent over $\overline{\mathbb{Q}}$. Here an infinite collection is independent when each of its finite subcollections is independent.

Proof. Take distinct odd primes $p_1, \dots, p_r$. For each j, choose a period vector $v_j \in \mathrm{Lie}(E_{p_j})_{\mathbb{C}}$ on which the invariant differential in (3.3) takes the value ℓ_{p_j}. In particular $v_j \neq 0$ and $\exp(v_j) = 0$. Suppose

$$c_0 + c_\pi \pi + \sum_{j=1}^{r} c_j \ell_{p_j} = 0, \qquad c_0, c_\pi, c_j \in \overline{\mathbb{Q}}.$$

Apply the analytic subgroup theorem to

$$G = \prod_{j=1}^{r} E_{p_j} \times \mathbb{G}_a \times \mathbb{G}_m, \qquad u = (v_1, \dots, v_r, 1, 2\pi i),$$

and to the invariant linear form with coefficients $c_1, \dots, c_r, c_0, c_\pi/(2i)$. It yields a connected subgroup H containing u in its Lie algebra and annihilated by that form.

The abelian and affine parts split as in Lemma 1.4. By Lemma 3.1, the elliptic factors are simple and pairwise nonisogenous. Hence each connected abelian subvariety of their product is a product of some of those factors: a nonproduct subvariety would yield a nonzero homomorphism between two distinct simple factors, and therefore an isogeny. Every projection of u to an elliptic Lie algebra is nonzero, so all elliptic factors must occur in H.

Likewise a connected subgroup of $\mathbb{G}_a \times \mathbb{G}_m$ splits between its additive and multiplicative factors. Its projections must both be nonzero since the corresponding coordinates of u are 1 and $2\pi i$. Thus $H = G$. The purported linear form therefore vanishes on all of $\mathrm{Lie}\, G$, forcing every coefficient to be zero. □

Corollary 3.3. *For distinct odd primes p, q, the quotient ℓ_p/ℓ_q is transcendental. For any finite choice of distinct odd primes, every nonzero algebraic linear combination of their perimeters is transcendental; adjoining 1 and π introduces no linear relation.*

Proof. An algebraic quotient would give a forbidden two-term linear relation. If a nonzero linear combination of the perimeters were algebraic, moving that algebraic number to the constant term would contradict Theorem 3.2. □

This is linear independence, not algebraic independence. No assertion about products of distinct perimeters, their polynomial relations, or the algebraic independence of π and any one perimeter is needed. The following register matches the numbered post-baseline results to the

precise statements here. Their hypotheses are retained, especially the noncircle restriction in entries 62–63 and the smooth-boundary restriction in entry 68.

61	Algebraic-endpoint arc lengths: Theorem 1.5 and the circle discussion following Corollary 1.6.
62	Noncircle length divided by π: Theorems 1.5 and 1.7.
63	Exponential, sine and cosine values for noncircle lengths: Corollary 1.8, also treating tangent.
64	Elementary antiderivative exactly for circles: Theorem 1.9.
65	Rational-lemniscate Euclidean and spherical lengths: Theorem 2.1.
66	Weighted lengths and even bending energies: Theorem 2.2.
67	Anisotropic length derivatives: Theorem 2.3.
68	Areas with smooth algebraic oval boundaries: Theorem 2.4.
69	Cassini perimeter independence: Theorem 3.2.

Part II
Division and inverse periods

4 Arithmetic division of real Abelian integrals

This section distinguishes three questions: which rational fractions of an Abelian integral have algebraic division points, whether replacing rational fractions by arbitrary algebraic fractions changes finiteness, and whether the resulting genus bounds can be improved. The answers are stated for holomorphic differentials on real algebraic curves. Consequently they apply to an arc-length differential only after its holomorphic realization on a smooth algebraic length cover has been established. A real oval means a connected component of the real locus of a smooth curve; it is a topological circle, not an elliptic curve.

Put $k = \overline{\mathbb{Q}} \cap \mathbb{R}$. Let C/k be a smooth projective geometrically connected curve of genus g, and let $0 \neq \omega \in H^0(C, \Omega^1_{C/k})$. We consider either of the following two settings.

1. A real oval $\Gamma \subset C(\mathbb{R})$ on which ω is nowhere zero, oriented so that its value on the positive tangent is positive. Fix $P_0 \in \Gamma(k)$ and traverse the oval once, returning to P_0.

2. An embedded compact arc $\Gamma \subset C(\mathbb{R})$ with distinct endpoints $P_0, P_1 \in C(k)$, oriented from P_0 to P_1, on which ω is positive and nowhere zero, including at the endpoints.

Let $L = \int_\Gamma \omega > 0$. For $0 \leq t \leq 1$, let P_t be the unique point reached after integral tL along the chosen traversal. In the oval case $P_1 = P_0$, and we count parameters in $I = [0, 1)$; in the arc case $I = [0, 1]$. Thus the identification at the end of a closed traversal is never counted twice. Define

$$S_{\mathbb{Q}} = \{t \in I \cap \mathbb{Q} : P_t \in C(k)\}, \qquad S_{\mathrm{alg}} = \{t \in I \cap k : P_t \in C(k)\}. \tag{4.1}$$

Positivity makes the parameter t single-valued. It also makes the traversal real analytic, because a local primitive of the nonvanishing differential is a real analytic coordinate.

4.1 The minimal Abelian quotient and a logarithmic rigidity lemma

Let $J = \mathrm{Jac}(C)$ and let $j : C \to J$ be the Abel–Jacobi map $P \mapsto [P - P_0]$. Write Ω for the invariant differential on J whose pullback is ω.

Lemma 4.1. *There is a unique largest Abelian subvariety $B \subset J_{\overline{\mathbb{Q}}}$ on which Ω vanishes. It is defined over k. The quotient*

$$\pi : J \longrightarrow A := J/B$$

and a nonzero invariant differential η on A are defined over k, with $\Omega = \pi^*\eta$. No positive-dimensional Abelian subvariety of $A_{\overline{\mathbb{Q}}}$ is annihilated by η. Moreover,

$$f := \pi \circ j : C \longrightarrow A$$

is nonconstant, its image contains 0 and generates A, and every fiber of f is finite.

Proof. The sum of two Abelian subvarieties annihilated by Ω has the same property, since its tangent space is the sum of their tangent spaces. Choose one of maximal dimension; taking its sum with any other proves both its maximality under inclusion and its uniqueness. Complex conjugation preserves the collection, and therefore preserves B. Descent to k, formation of the quotient, and descent of invariant differentials give A and η. If $D \subset A$ were a positive-dimensional Abelian subvariety annihilated by η, the connected inverse image of D in J would strictly contain B and would be annihilated by Ω. This is impossible.

Since $f^*\eta = \omega \neq 0$, the map f is nonconstant. The image of j generates J, so $f(C)$ generates A. A nonconstant morphism from a smooth projective integral curve has finite fibers; otherwise a fiber would contain the entire curve. These observations also show that $\dim A \geq 1$. □

We call A the *minimal Abelian quotient of* (C, ω). Its dimension, rather than the genus alone or the decomposition of J alone, governs arithmetic division. In particular, a split Jacobian does not by itself imply that ω comes from an elliptic curve.

Our transcendence input is the following consequence of the analytic subgroup theorem in its logarithmic form [2, Theorem 6.2]. If $G/\overline{\mathbb{Q}}$ is a connected commutative algebraic group, $u \in \mathrm{Lie}(G)_{\mathbb{C}}$, $\exp_G(u) \in G(\overline{\mathbb{Q}})$, and $V \subset \mathrm{Lie}(G)$ is a $\overline{\mathbb{Q}}$-linear subspace with $u \in V_{\mathbb{C}}$, then there is a connected algebraic subgroup $H \subset G$, defined over $\overline{\mathbb{Q}}$, such that

$$u \in \mathrm{Lie}(H)_{\mathbb{C}} \subset V_{\mathbb{C}}. \tag{4.2}$$

The assertion includes the case of a nonzero logarithm whose exponential is the identity. It is therefore applicable to periods, without an extra non-torsion hypothesis.

Lemma 4.2 (Rigidity in the kernel)**.** *If $w \in \mathrm{Lie}(A)_{\mathbb{C}}$ satisfies $\exp_A(w) \in A(\overline{\mathbb{Q}})$ and $\eta(w) = 0$, then $w = 0$.*

Proof. Apply (4.2) with $V = \ker \eta$. A nonzero w would produce a positive-dimensional Abelian subvariety of A annihilated by η, contrary to Lemma 4.1. □

Lift the path $f(P_t)$ to a continuous logarithm $u_t \in \mathrm{Lie}(A)_{\mathbb{C}}$ with $u_0 = 0$, and put $v = u_1$. Integration of invariant differentials gives

$$\exp_A(u_t) = f(P_t), \quad \exp_A(v) = f(P_1), \quad \eta(u_t) = tL, \quad \eta(v) = L. \tag{4.3}$$

In the oval case v is a period. In the arc case it is a logarithm of an algebraic point; the proofs below do not replace it by a period.

Lemma 4.3 (Rational division forces vector division)**.** *If $q \in I \cap \mathbb{Q}$ and P_q is algebraic, then $u_q = qv$. Conversely, if $q \in I \cap \mathbb{Q}$ and $u_q = qv$, then P_q is algebraic.*

Proof. For $q = m/n$, with $n > 0$, the point $\exp_A(qv)$ is algebraic: its n-multiple equals the algebraic point $[m]f(P_1)$, and $[n] : A \to A$ is finite. If P_q is algebraic, then $u_q - qv$ has algebraic exponential and is annihilated by η. Lemma 4.2 gives the required equality. Conversely, the equality makes $f(P_q)$ algebraic. Its finite fiber under the morphism f, which is defined over $\overline{\mathbb{Q}}$, consists of algebraic points. □

4.2 The rational division dichotomy and the genus bounds

Theorem 4.4 (Rational division dichotomy)**.** *Under the preceding hypotheses, the following conditions are equivalent:*

1. $S_{\mathbb{Q}}$ *is infinite;*
2. $S_{\mathbb{Q}} = I \cap \mathbb{Q}$*;*
3. $\dim A = 1$*;*
4. *there are an elliptic curve* $E/\overline{\mathbb{Q}}$*, a nonconstant morphism* $h : C_{\overline{\mathbb{Q}}} \to E$*, and a nonzero invariant differential* ν *on* E *such that* $\omega = h^*\nu$*.*

If these conditions do not hold, then

$$\#S_{\mathbb{Q}} \leq \begin{cases} 2g-2, & \text{for an oval, with } q=0 \text{ included;} \\ 2g-1, & \text{for an arc, with both endpoints included.} \end{cases} \tag{4.4}$$

Proof. If $\dim A = 1$, the nonzero functional η has zero kernel. Equations (4.3) therefore imply $u_q = qv$ for every real q, and Lemma 4.3 proves algebraicity for every rational q.

Suppose instead that $\dim A > 1$. Choose a real invariant differential ξ on A, defined over k, independent of η, and put $\theta = f^*\xi$. The pullback on invariant differentials is injective: it is the composite of the injective pullback from A to J and the Abel–Jacobi identification with $H^0(C, \Omega^1)$. Hence θ and ω are independent. Set

$$c = \frac{\int_\Gamma \theta}{L} \in \mathbb{R}, \qquad \delta = \theta - c\omega, \qquad H(t) = \int_{P_0}^{P_t} \delta.$$

Here the path of integration is the initial part of the fixed traversal. The differential δ is a nonzero holomorphic differential on $C_{\mathbb{C}}$, although c need not be algebraic. Consequently its zero divisor has degree $2g-2$. The function H is real analytic, real valued, and satisfies $H(0) = H(1) = 0$. Differentiating in the normalized integral coordinate gives

$$H'(t) = L\,\frac{\delta}{\omega}(P_t). \tag{4.5}$$

Thus the derivative has at most $2g-2$ zeros on a simple traversal. The function H is not identically zero, since otherwise δ would vanish on a real arc and hence on C.

For every $q \in S_{\mathbb{Q}}$, Lemma 4.3 implies

$$\int_{P_0}^{P_q} \theta = q \int_\Gamma \theta,$$

so $H(q) = 0$. In the arc case, N distinct zeros of H, including the endpoints, require at least $N-1$ distinct zeros of H' by Rolle's theorem. This gives $N \leq 2g-1$. In the oval case H is a real analytic function on the parameter circle: its total increment is zero, and the periodic differential determines its periodic analytic extension. Cyclic Rolle's theorem gives at least N derivative zeros from N distinct zeros of H on the circle. Therefore $N \leq 2g-2$. The same arguments applied to any finite subset exclude an infinite zero set and prove (4.4).

This establishes the equivalence of the first three conditions. The third implies the fourth by taking $E = A$ and $h = f$. Conversely, translate h so that $h(P_0) = 0$. The identity component of the kernel of the induced surjection $J \to E$ has codimension one and is annihilated by Ω. Lemma 4.1 then gives $\dim A \leq 1$, and it is nonzero. □

Corollary 4.5 (A finite elliptic-reduction certificate)**.** *On an oval, the existence of more than* $2g-2$ *distinct algebraic rational-division points forces elliptic reduction of the differential and hence algebraicity of every rational-division point. On an arc, the corresponding threshold is more than* $2g-1$ *points, with its two endpoints counted.*

For clarity, the open-arc bound permits at most $2g-3$ internal algebraic rational-division points: its two algebraic endpoints already account for two of the $2g-1$ points. The closed-oval bound counts $q=0$ once and does not count the duplicate parameter $q=1$.

Remark 4.6. When these results are applied to polynomial lemniscates, an oval is an oval on the smooth length cover. A regular geometric component admits such a traversal. At a critical level, a geometric component may be a graph, and a route switching analytic branches can jump between different points of the normalization. Such a piecewise route is not automatically an oval or an arc in the present theorem. The theorem also concerns a single chosen traversal, not an unordered sum of component lengths.

4.3 Algebraic fractions and endomorphisms

Proposition 4.7 (Rationality under scalar endomorphisms)**.** *Suppose* $\mathrm{End}^0_{\overline{\mathbb{Q}}}(A)=\mathbb{Q}$*. If* $q\in S_{\mathrm{alg}}$*, then* $q\in\mathbb{Q}$ *and* $u_q=qv$*. Consequently* $S_{\mathrm{alg}}=S_{\mathbb{Q}}$*, and for* $\dim A>1$ *the bounds* (4.4) *hold for all algebraic fractions.*

Proof. The endpoints cause no difficulty, so take $0<q<1$. The hypothesis implies that A is absolutely simple: a nontrivial isogeny decomposition would give a nontrivial idempotent in its rational endomorphism algebra. Apply (4.2) to (u_q,v) on A^2 and the algebraic hyperplane

$$V_q=\{(U,V):\eta(U)-q\eta(V)=0\}.$$

The resulting connected subgroup H is proper. Both projections of H to A are nonzero, since u_q and v are nonzero, and are thus surjective. Simplicity shows that H has dimension $\dim A$ and each projection is an isogeny. As $\mathrm{End}^0(A)=\mathbb{Q}$, there are nonzero integers m,n such that the Lie algebra of H is the image of $W\mapsto(mW,nW)$. Its containment in V_q forces $m-qn=0$. Hence $q=m/n$ and $u_q=qv$. □

Proposition 4.8 (The exact elliptic alternative)**.** *If* $\dim A=1$*, then* $S_{\mathrm{alg}}=I\cap\mathbb{Q}$*, whether or not* A *has complex multiplication.*

Proof. Rational fractions are already covered by Theorem 4.4. For an algebraic interior fraction with algebraic division point, apply the preceding A^2 argument. A proper connected subgroup with both projections nonzero is the graph of a quasi-endomorphism in the isogeny category. Thus q is the analytic scalar of a member of $\mathrm{End}^0(A)$. This algebra is either $\mathbb{Q}$ or an imaginary quadratic field. Its real analytic scalars are precisely $\mathbb{Q}$, proving the assertion. □

The scalar hypothesis in Proposition 4.7 should not be dropped from its proof. For example, on a simple Abelian variety with real multiplication, a chosen differential may be an eigenvector for irrational algebraic scalars. The vector relation supplied by the subgroup theorem can then involve a non-scalar endomorphism. The following result gives the general finiteness statement without pretending that this vector relation is scalar.

Lemma 4.9 (Reduction to a division hull)**.** *Let* $Q=f(P_1)$ *and*

$$\mathcal{G}=\mathrm{End}_{\overline{\mathbb{Q}}}(A)\cdot Q=\{\phi(Q):\phi\in\mathrm{End}_{\overline{\mathbb{Q}}}(A)\}.$$

This is a finitely generated subgroup of $A(\overline{\mathbb{Q}})$*. If* $q\in S_{\mathrm{alg}}$*, then*

$$[N]f(P_q)=\phi(Q)\quad\text{for some } N\geq 1 \text{ and } \phi\in\mathrm{End}_{\overline{\mathbb{Q}}}(A). \tag{4.6}$$

In particular, if Q is torsion, then $f(P_q)$ is torsion.

Proof. The endomorphism ring of an Abelian variety is a finitely generated abelian group, so its image $\mathcal{G}$ is finitely generated. Apply (4.2) to (u_q, v) and V_q as above, without any simplicity assumption, and let $H \subset A^2$ be the resulting connected Abelian subvariety. The identity component of the kernel of $p_2 := \mathrm{pr}_2 |_H$ lies in $A \times \{0\}$ and is annihilated by η. By the defining property of A, this identity component is zero. Therefore $p_2 : H \to D := p_2(H)$ is an isogeny.

In the category of Abelian varieties up to isogeny, put $\psi = p_1 p_2^{-1} \in \mathrm{Hom}^0(D, A)$. Poincaré complete reducibility supplies a rational retraction $r : A \to D$ whose restriction to D is the identity. The quasi-endomorphism $\Phi = \psi r \in \mathrm{End}^0(A)$ satisfies $p_1 = \Phi p_2$ on H in this category. Clear all denominators to obtain $N \geq 1$ and an actual endomorphism $\phi = N\Phi$ such that $[N]p_1 = \phi p_2$ as homomorphisms on H. Since $(u_q, v) \in \mathrm{Lie}(H)_{\mathbb{C}}$, its exponential $(f(P_q), Q)$ belongs to H. Evaluating the last identity yields (4.6). The final assertion follows immediately. □

For a subgroup $\mathcal{G}$ of an Abelian variety, write

$$\mathrm{Div}(\mathcal{G}) = \{R : [N]R \in \mathcal{G} \text{ for some } N \geq 1\}.$$

We use the following classical curve case of the division-point theorem: if X is an integral nonelliptic projective curve in an Abelian variety in characteristic zero and $\mathcal{G}$ is a finitely generated subgroup, then $X(\overline{K}) \cap \mathrm{Div}(\mathcal{G})$ is finite. The statement, including passage from a finitely generated group to its division hull, is Theorem 1 of [7], following Raynaud and Faltings. We use this curve theorem, rather than inferring division-hull finiteness merely from finite generation of $\mathcal{G}$.

Theorem 4.10 (Dichotomy for all algebraic fractions)**.** *If $\dim A > 1$, then S_{alg} is finite. If $\dim A = 1$, then $S_{\mathrm{alg}} = I \cap \mathbb{Q}$. Thus the following are equivalent: infinitely many algebraic fractions have algebraic division points; every rational fraction has an algebraic division point; and the differential comes from an elliptic curve.*

Proof. The elliptic case is Proposition 4.8. Suppose $\dim A > 1$ and put $X = f(C)$. This is an integral projective curve. It is nonelliptic: a genus-one curve in an Abelian variety is a translate of an elliptic subgroup, and if that curve contains 0 it is the subgroup itself. Such an X could not generate an Abelian variety of dimension greater than one.

Lemma 4.9 places the images of all points counted by S_{alg} in $X \cap \mathrm{Div}(\mathcal{G})$, a finite set by the preceding division-point theorem. The map f has finite fibers, and our traversal is injective, with its closed endpoint identified. Hence S_{alg} is finite. □

Remark 4.11. Theorem 4.10 does not provide an effective list of the exceptional algebraic fractions, nor the numerical bounds (4.4) without an additional scalar-endomorphism hypothesis. In the closed-oval case $Q = 0$, its finiteness step is already the Manin–Mumford theorem. The division-hull formulation is what also treats the open arc and non-torsion algebraic endpoints.

4.4 Optimality of the closed-oval bound in every genus

Theorem 4.12 (Sharpness)**.** *For every integer $g \geq 2$, there are data (C, ω, Γ, P_0) in the oval setting above for which the minimal Abelian quotient has dimension two and*

$$S_{\mathrm{alg}} = S_{\mathbb{Q}} = \left\{ \frac{k}{2g-2} : 0 \leq k < 2g-2 \right\}. \tag{4.7}$$

Thus $2g-2$ in (4.4) is optimal in every genus, even if all real algebraic fractions are allowed.

Proof. We give the algebraic base curve, the covering construction, and its descent explicitly enough to specify an existence proof over k. Start with the smooth projective model

$$C_0 : \quad y^2 = F(x), \qquad F(x) = x^5 - 6x^3 + 5x + 1. \tag{4.8}$$

The signs at $-3, -2, -1/2, 1/2, 2, 3$ alternate as $-, +, -, +, -, +$. There are consequently five distinct real roots, denoted $a_1 < \cdots < a_5$. The curve has genus two and three real ovals. In particular, the finite oval over $[a_1, a_2]$ carries the positive nowhere-zero differential $\omega_0 = dx/y$, with the orientation going from a_1 to a_2 on the positive sheet.

We verify the arithmetic of its Jacobian. Modulo 2,

$$F = (x^2 + x + 1)(x^3 + x^2 + 1),$$

with both factors irreducible. Modulo 3, $F = x^5 - x + 1$ is irreducible: it has no roots, and its remainders on division by the three monic irreducible quadratics x^2+1, x^2+x+2, x^2+2x+2 are 1, $x+1$, $x+1$, respectively. Both reductions are squarefree. Their Frobenius cycle types give a 5-cycle and an element of type (2)(3) in the Galois group; cubing the latter gives a transposition. A 5-cycle and any transposition generate S_5: the conjugate transpositions form the edge set of a connected cyclic graph on five vertices. Hence $\mathrm{Gal}(F) = S_5$, and [8, Theorem 2.1] gives

$$\mathrm{End}_{\overline{\mathbb{Q}}} J(C_0) = \mathbb{Z}. \tag{4.9}$$

In particular $J(C_0)$ is absolutely simple, and the minimal quotient for ω_0 is the whole two-dimensional Jacobian.

Let $d = g - 1$. The quotient of $C_0(\mathbb{C})$ by conjugation is a pair of pants S, namely a sphere with three boundary circles. One can see this directly from the three real ovals and the Euler characteristic: the quotient has three boundary components and Euler characteristic -1, which forces it to be an orientable genus-zero surface. The original Riemann surface is its double.

Label the boundary corresponding to the selected finite oval first. The fundamental group of S has boundary generators with product 1. Send these three generators to $1, 0, -1$ in $\mathbb{Z}/d\mathbb{Z}$. This defines a connected cyclic cover $S_d \to S$ of degree d. There is one boundary circle over the first boundary, d over the second, and one over the third; the first and third are degree-d covers. Since $\chi(S_d) = -d$, its $d + 2$ boundary components force its genus to be zero. Pull back the bordered complex structure and double along the analytic boundaries. The resulting compact Riemann surface C_d has an antiholomorphic involution and a holomorphic unramified map

$$\rho : C_d \longrightarrow C_0$$

of degree d. There is no ramification along a doubled boundary: in a local half-disc the covering is a local biholomorphism, and reflection extends it across the boundary. Riemann–Hurwitz gives $g(C_d) = d + 1 = g$. The inverse image of the chosen oval is a single real oval Γ_d mapping onto it with degree d.

The construction is algebraic over k, not merely over $\mathbb{C}$. Indeed, Riemann existence algebraizes the finite unramified cover. The equivalence of finite étale cover categories under extension of algebraically closed fields of characteristic zero descends it from $\mathbb{C}$ to $\overline{\mathbb{Q}}$; see [9, Remark 3.3(c) and Theorem 3.4]. The reflection is an isomorphism from the conjugate cover to the cover over the real base C_0. Full faithfulness of the same equivalence makes this isomorphism defined over $\overline{\mathbb{Q}}$. Its square is the identity, so it is a Galois descent datum for $\overline{\mathbb{Q}}/k$. Effective descent for finite étale covers yields C_d, ρ, and their real structure over k.

Set $\omega_d = \rho^*\omega_0$ and take a point above $(a_1, 0)$ as the algebraic basepoint. The norm homomorphism $\rho_* : J(C_d) \to J(C_0)$ is surjective. Its connected kernel is exactly the largest Abelian subvariety annihilated by the invariant differential corresponding to ω_d. To check maximality,

any annihilated Abelian subvariety maps either to zero or onto the simple $J(C_0)$. The latter would force the nonzero differential corresponding to ω_0 to vanish identically. It follows that the minimal quotient is isogenous to $J(C_0)$, has dimension two, and has rational endomorphism algebra $\mathbb{Q}$.

The two halves of the base oval have equal positive ω_0 integrals, by the hyperelliptic involution. The d points above each endpoint $(a_1, 0)$ and $(a_2, 0)$ are all on Γ_d, are algebraic, and occur in alternating order at integral fractions $k/(2d)$ of its total integral. Thus there are at least $2d$ points in $S_{\mathbb{Q}}$. Theorem 4.4 permits at most $2g - 2 = 2d$. Proposition 4.7 excludes additional irrational algebraic fractions. This proves (4.7). □

Remark 4.13. The construction proves sharpness for holomorphic Abelian integrals on real ovals. It does not assert that the curves C_d are length covers of polynomial lemniscates, nor supply explicit plane equations for every d. The field of definition and the absence of ramification are part of the proof, rather than assumptions about a topological cover. In particular the construction works for both odd and even d.

Example 4.14 (The open-arc constant in genus two)**.** On a finite oval of the base curve (4.8), choose an algebraic nonbranch point P and its hyperelliptic conjugate $\iota(P)$. The simple arc between them that passes through one branch point has that branch point at exactly half its integral. The endpoints and midpoint are therefore three algebraic rational division points. Since $g = 2$ and the minimal quotient has dimension two, the bound $2g - 1 = 3$ excludes all others; scalar endomorphisms also exclude irrational algebraic fractions. This proves sharpness of the open-arc bound in genus two. No all-genus sharpness assertion for open arcs is made here.

5 A complete rational-quantile classification for the beta family

This section classifies rational quantiles of $\mathrm{Beta}(\alpha, 1/2)$ for every positive rational α. The statement includes conditioning on any nondegenerate interval with algebraic endpoints. Its geometric form classifies division points on monotone half-leaves of all sinusoidal spirals with positive rational parameter. The proof uses the analytic subgroup theorem and a cyclotomic component of a generalized Jacobian. In particular, it does not require that the relevant abelian variety be simple.

5.1 The classification and its integral form

For $\alpha > 0$ and $0 \le u < v \le 1$, write

$$F_{\alpha;u,v}(x) = \frac{\displaystyle\int_u^x t^{\alpha-1}(1-t)^{-1/2}\,dt}{\displaystyle\int_u^v t^{\alpha-1}(1-t)^{-1/2}\,dt}, \qquad u \le x \le v.$$

All endpoint integrals in this section are understood as convergent improper integrals when necessary. The function $F_{\alpha;u,v}$ is continuous and strictly increasing from 0 to 1.

Theorem 5.1 (Rational beta quantiles; result 76)**.** *Let $\alpha \in \mathbb{Q}_{>0}$, let $u, v \in \overline{\mathbb{Q}} \cap \mathbb{R}$ satisfy $0 \le u < v \le 1$, and let $r \in \mathbb{Q} \cap (0, 1)$. There is a unique $x \in (u, v)$ with $F_{\alpha;u,v}(x) = r$, and*

$$x \in \overline{\mathbb{Q}} \quad \Longleftrightarrow \quad \alpha \in \mathbb{Z}_{>0} \cup \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}\right\}. \tag{5.1.1}$$

Consequently, outside this exceptional parameter set, every internal rational quantile is transcendental, including every such quantile of every algebraically truncated distribution in the family.

The equivalent integral statement is more convenient for the proof. For coprime positive integers a, b, put

$$J_{a,b}(s) = \int_0^s \frac{t^{b-1}}{\sqrt{1-t^{2a}}}\,dt, \qquad 0 \le s \le 1.$$

Define

$$\mathcal{E} = \{(1,1),(2,1),(3,1),(3,2)\} \ \cup \ \{(1,2k) : k \ge 1\}.$$

Theorem 5.2 (Monomial-weighted hyperelliptic division)**.** *Let $(a,b) = 1$, let $0 \le u < v \le 1$ be algebraic, and let $r \in \mathbb{Q} \cap (0,1)$. The unique $s \in (u,v)$ satisfying*

$$J_{a,b}(s) - J_{a,b}(u) = r\big(J_{a,b}(v) - J_{a,b}(u)\big) \tag{5.1}$$

is algebraic if $(a,b) \in \mathcal{E}$, and transcendental otherwise.

The change of variables $x = s^{2a}$ gives the exact identity

$$J_{a,b}(s) = \frac{1}{2a} B_{s^{2a}}\left(\frac{b}{2a}, \frac{1}{2}\right). \tag{5.2}$$

For every $\alpha \in \mathbb{Q}_{>0}$, the fraction $1/(2\alpha) = a/b$ has a unique expression in coprime positive integers. Positive real algebraicity is preserved in both directions by $s \mapsto s^{2a}$, including at interval endpoints. The members of $\mathcal{E}$ correspond precisely to the parameter set in Theorem 5.1. Thus it suffices to prove Theorem 5.2.

5.2 Two inputs and a strict comparison

We use the following form of the analytic subgroup theorem. If G is a connected commutative algebraic group over $\overline{\mathbb{Q}}$, $z \in \mathrm{Lie}(G) \otimes_{\overline{\mathbb{Q}}} \mathbb{C}$, and $\exp_G(z) \in G(\overline{\mathbb{Q}})$, then there is a connected algebraic subgroup $H \subseteq G$ such that

$$z \in \mathrm{Lie}(H)_{\mathbb{C}}, \qquad \mathrm{Lie}(H) \subseteq \ker \ell$$

for every algebraic linear functional ℓ with $\ell(z) = 0$. The case $z = 0$ permits $H = 0$. This is the canonical subgroup formulation of Wüstholz's theorem given in [2, Theorem 6.2].

We also use the standard generalized Jacobian of a smooth projective curve C with an effective modulus $\mathfrak{m}$. If S is the support of $\mathfrak{m}$ and $P_0 \notin S$, its generalized Abel map

$$j_{\mathfrak{m}} : C \setminus S \longrightarrow J_{\mathfrak{m}}, \qquad P \longmapsto [P - P_0]_{\mathfrak{m}},$$

is algebraic. Pullback identifies invariant differentials on $J_{\mathfrak{m}}$ with meromorphic differentials on C whose poles are bounded by $\mathfrak{m}$. In characteristic zero, the connected affine kernel of the map to the ordinary Jacobian has a torus of dimension $|S| - 1$ and a vector-group part. These facts, including functoriality under automorphisms preserving the modulus, are due to Rosenlicht and Serre; see [10] and [11, Chapter V]. A vector group has contractible complex points, so passing from a vector extension to its abelian quotient induces an isomorphism on rational first homology.

Lemma 5.3 (Strict comparison of weighted fractions)**.** *Let $0 \le u < s < v \le 1$ and a, b, k be positive integers with $b \ne k$. For $j = b, k$ define*

$$R_j(s) = \frac{\displaystyle\int_u^s t^{j-1}(1-t^{2a})^{-1/2}\,dt}{\displaystyle\int_u^v t^{j-1}(1-t^{2a})^{-1/2}\,dt}.$$

If $b > k$, then $R_b(s) < R_k(s)$; if $b < k$, the reverse inequality holds.

Proof. Use $t^{k-1}(1-t^{2a})^{-1/2}dt$ as a positive reference measure. When $b > k$, the multiplier t^{b-k} is strictly increasing. Its mean on (u,s) is strictly smaller than its mean on (s,v), since both intervals have positive measure. Multiplying out these two means gives $R_b(s) < R_k(s)$. Interchanging b and k proves the other case. The argument remains valid when $u = 0$ or $v = 1$, because all the measures in question have finite mass. □

5.3 The cyclotomic component and its periods

Assume throughout this subsection that $a \geq 2$ and $(a,b) = 1$. Let C_a be the smooth projective model of

$$y^2 = 1 - x^{2a},$$

and set

$$\zeta = \exp(2\pi i/(2a)), \qquad \tau(x,y) = (\zeta x, y), \qquad \Omega_b = \frac{x^{b-1}\,dx}{y}.$$

The genus is $a-1$, and its holomorphic differentials have basis

$$\omega_{k-1} = \frac{x^{k-1}\,dx}{y}, \qquad 1 \leq k < a, \qquad \tau^*\omega_{k-1} = \zeta^k \omega_{k-1}. \tag{5.3}$$

Together with their antiholomorphic conjugates, these characters show that the characteristic polynomial of τ_* on $H_1(C_a, \mathbb{Q})$ is

$$\frac{T^{2a}-1}{T^2-1}. \tag{5.4}$$

In particular, every cyclotomic factor occurring here has multiplicity one. Put

$$g = \gcd(b, 2a), \qquad d = \frac{2a}{g}.$$

Coprimality of a, b implies $g \in \{1,2\}$ and $d \geq 3$.

Lemma 5.4 (A generalized cyclotomic component)**.** *There exist a connected commutative algebraic group G over $\overline{\mathbb{Q}}$, an algebraic map $j : C_a \setminus \{\infty_+, \infty_-\} \to G$, and an invariant differential Θ_b on G with the following properties.*

1. *There is an exact sequence*

$$0 \longrightarrow U \longrightarrow G \xrightarrow{p} A \longrightarrow 0,$$

   where U is a vector group and A is an abelian variety.

2. *$j(0,1) = 0$ and $j^*\Theta_b = \Omega_b$.*

3. *The space $H_1(G,\mathbb{Q}) \cong H_1(A,\mathbb{Q})$ is an irreducible $\mathbb{Q}[T]$-module with characteristic polynomial $\Phi_d(T)$, where T acts by the automorphism induced by τ.*

4. *Every k with $1 \leq k < a$ and $\gcd(k,2a) = g$ yields an invariant differential Θ_k pulled back from A such that $j^*\Theta_k = \omega_{k-1}$.*

5. *The period map*

$$H_1(G,\mathbb{Q}) \longrightarrow \mathbb{C}, \qquad \gamma \longmapsto \int_\gamma \Theta_b$$

   is injective.

Proof. The two points at infinity are exchanged by τ. Indeed, in the coordinates $t = 1/x$ and $w = y/x^a$, the two limiting values of w have opposite signs, and τ sends w to $-w$. All finite points are regular for Ω_b: at a branch point one may use y as a local parameter. At infinity its leading order is $t^{a-b-1}dt$. Choose an integer $M \geq \max\{1, b-a+1\}$ and take $\mathfrak{m} = M(\infty_+ + \infty_-)$. This modulus is τ-invariant and bounds the poles of Ω_b.

Set $J^{\mathfrak{m}} = J_{\mathfrak{m}}(C_a)$, pointed at $(0,1)$. The latter point is fixed by τ, so τ induces an automorphism T of $J^{\mathfrak{m}}$ of order dividing $2a$, and its Abel map is equivariant. The connected affine part of $J^{\mathfrak{m}}$ has torus rank one. On its torus, T acts by inversion because it exchanges the two points in the support of the modulus. Thus this torus has character -1, belonging to Φ_2.

Let $e_d \in \mathbb{Q}[T]/(T^{2a}-1)$ be the idempotent which is 1 on the Φ_d-component and 0 on all other cyclotomic components. Choose a positive integer N clearing its denominators, and let

$$f = Ne_d \in \mathrm{End}(J^{\mathfrak{m}}), \qquad G = \mathrm{im}(f).$$

Images here are taken as algebraic subgroups; G is connected. The Φ_2 torus has been killed. More explicitly, $f^2 = Nf$, so f restricts to multiplication by N on G. Any torus in G is also a torus in $J^{\mathfrak{m}}$, where f vanishes; it must therefore be trivial. Thus the connected affine part of G is a vector group. Let A be the actual Chevalley abelian quotient of G. It is isogenous to the Φ_d-factor of $\mathrm{Jac}(C_a)$. Using the Chevalley quotient avoids assuming that the map from G to its image in the ordinary Jacobian has connected kernel.

The invariant differential on $J^{\mathfrak{m}}$ pulling back to Ω_b is a T^*-eigenvector of eigenvalue ζ^b, which has order d. Write it as $\widetilde{\Omega}_b$. If $\iota : G \hookrightarrow J^{\mathfrak{m}}$ is inclusion, define

$$j = f \circ j_{\mathfrak{m}}, \qquad \Theta_b = N^{-1}\iota^*\widetilde{\Omega}_b.$$

As $e_d^*\widetilde{\Omega}_b = \widetilde{\Omega}_b$,

$$j^*\Theta_b = N^{-1}j_{\mathfrak{m}}^* f^*\widetilde{\Omega}_b = \Omega_b.$$

This normalization accounts for the integer used to clear denominators. The identical construction applies to every differential in (5.3) of character order d. Those differentials come from the ordinary Jacobian, vanish on every affine direction, and hence factor through A.

The rational first homology of $J^{\mathfrak{m}}$ is that of C_a with the two points at infinity removed. The extra homology class is the toric class of character -1. Consequently its Φ_d-part is the multiplicity-one Φ_d-part in (5.4). On rational homology the image of f is precisely this part, since the scalar N is invertible over $\mathbb{Q}$. One may also see this from analytic period lattices: inclusion of a connected algebraic subgroup induces an injective map on first homology, and a surjective homomorphism induces a surjection on rational first homology. This proves the asserted irreducibility and dimension $\varphi(d)$.

It remains to verify that the period map is nonzero. Let δ be the path from $(0,1)$ to $(1,0)$ on the positive square-root sheet, followed by the path from $(1,0)$ to $(0,-1)$ on the negative sheet. Then $\delta - \tau\delta$ is a closed cycle away from infinity, and

$$\int_{\delta-\tau\delta} \Omega_b = 2(1-\zeta^b)J_{a,b}(1) \neq 0.$$

Its image under j gives a nonzero period of Θ_b. The kernel of the period map is T-stable, because

$$\int_{T_*\gamma} \Theta_b = \zeta^b \int_\gamma \Theta_b.$$

Irreducibility over $\mathbb{Q}$ now implies that this kernel is zero. □

Lemma 5.5 (Vanishing forces the abelian logarithm to vanish)**.** *In the setting of Lemma 5.4, suppose* $z \in \mathrm{Lie}(G)_{\mathbb{C}}$, $\exp_G(z) \in G(\overline{\mathbb{Q}})$, *and* $\Theta_b(z) = 0$. *Then* $dp(z) = 0$ *in* $\mathrm{Lie}(A)_{\mathbb{C}}$.

Proof. The analytic subgroup theorem gives a connected algebraic subgroup $H \subseteq G$ with $z \in \mathrm{Lie}(H)_{\mathbb{C}}$ and $\mathrm{Lie}(H) \subseteq \ker \Theta_b$. Let $B = p(H) \subseteq A$. It is an abelian subvariety. The kernel of $H \to B$ is $H \cap U$, an algebraic subgroup of a vector group and therefore itself a vector group in characteristic zero. It is connected and analytically contractible. Hence

$$H_1(H, \mathbb{Q}) \longrightarrow H_1(B, \mathbb{Q})$$

is an isomorphism. If $\dim B > 0$, inclusion of B in A supplies a nonzero image in $H_1(A, \mathbb{Q})$ and thus in $H_1(G, \mathbb{Q})$. Its Θ_b-period is zero, since the restriction of this invariant differential to H is zero. This contradicts the injectivity in Lemma 5.4. Thus $B = 0$, whence $H \subseteq U$ and $dp(z) = 0$. □

5.4 Proof of the classification

Proof of Theorem 5.2. First suppose $a \geq 2$. For $t \in [0, 1]$, put $P_t = (t, \sqrt{1 - t^{2a}})$, and choose the logarithm z_t of $j(P_t)$ obtained by integrating along the real path from P_0 to P_t. The normalization in Lemma 5.4 gives

$$\Theta_b(z_t) = J_{a,b}(t), \qquad \Theta_k(z_t) = J_{a,k}(t)$$

for all the holomorphic indices k occurring in that lemma.

Suppose that the solution s of (5.1) is algebraic, and write $r = m/h$ with positive integers $0 < m < h$. All $j(P_t)$ for $t = u, s, v$ are algebraic points. Therefore

$$z = h(z_s - z_u) - m(z_v - z_u)$$

has algebraic exponential, and (5.1) states $\Theta_b(z) = 0$. Lemma 5.5 gives $dp(z) = 0$. Every holomorphic differential Θ_k in the same cyclotomic component consequently satisfies $\Theta_k(z) = 0$. In other words,

$$\frac{J_{a,k}(s) - J_{a,k}(u)}{J_{a,k}(v) - J_{a,k}(u)} = r. \tag{5.5}$$

If one such index k differs from b, this contradicts Lemma 5.3.

If $b > a$, take $k = g = \gcd(b, 2a)$. For odd b this is $k = 1 < a$; for even b, the integer a is odd and at least 3, so $k = 2 < a$. In either case $k < b$ and the character order is d. This proves transcendence in every case with $a \geq 2$ and $b > a$, including cases where Ω_b is of the second kind.

If $b < a$, there are exactly $\varphi(d)/2$ indices k with $1 \leq k < a$ and $\gcd(k, 2a) = g$. When $\varphi(d) \geq 4$, one of them differs from b, again giving a contradiction. The only remaining possibility is $\varphi(d) = 2$, namely $d \in \{3, 4, 6\}$. Under $b < a$ and $(a, b) = 1$, these cases are exactly

$$(a, b) = (2, 1), (3, 1), (3, 2).$$

For $(2, 1)$ the curve $y^2 = 1 - x^4$ has genus one and the length differential is a nonzero holomorphic differential. For $(3, 2)$ the map

$$(x, y) \longmapsto (X, Y) = (x^2, y), \qquad Y^2 = 1 - X^3,$$

satisfies $dX/Y = 2x\, dx/y$. For $(3, 1)$ use instead

$$(x, y) \longmapsto (X, Y) = (x^{-2}, iy/x^3), \qquad dX/Y = 2i\, dx/y.$$

Both rational maps extend to nonconstant morphisms between the smooth projective curves. In particular the point $x = 0$ in the latter map is interpreted on the projective models. Thus in each case Ω_b is the pullback of a nonzero invariant differential on an elliptic curve, up to a nonzero algebraic scalar.

On that elliptic curve the equality of scalar integrals in (5.1) is an equality of logarithms, because the Lie algebra is one-dimensional. It follows that h times the image of P_s is an algebraic group combination of the images of P_u and P_v. Multiplication by h has finite algebraic fibers, so the image of P_s is algebraic. A nonconstant morphism of smooth projective curves is finite, hence P_s, and then s, is algebraic. This proves all three genus-one exceptions without requiring the endpoints to be torsion points.

It remains to consider $a = 1$. If $b = 2k$ is even, the substitution $y = \sqrt{1 - s^2}$ gives

$$J_{1,2k}(s) = \int_y^1 (1 - t^2)^{k-1}\, dt.$$

This is a rational-coefficient polynomial in y. For algebraic endpoints, the division equation is a nonconstant polynomial equation over $\overline{\mathbb{Q}}$, so its solution y, and therefore s, is algebraic. The same argument allows any algebraic $r \in (0, 1)$.

For $b = 1$, write $\theta_t = \arcsin t \in [0, \pi/2]$. Then $J_{1,1}(t) = \theta_t$. At an algebraic t,

$$e^{i\theta_t} = \sqrt{1 - t^2} + it$$

is algebraic. The relation $h(\theta_s - \theta_u) = m(\theta_v - \theta_u)$ shows that $e^{i\theta_s}$ is an h-th root of an algebraic number, and hence s is algebraic.

Finally let $b = 2k + 1 \geq 3$. Repeated use of

$$J_{1,b}(t) = \frac{b-2}{b-1} J_{1,b-2}(t) - \frac{t^{b-2}\sqrt{1 - t^2}}{b-1} \tag{5.6}$$

gives

$$J_{1,b}(t) = c_k \theta_t + R(t), \qquad c_k = 4^{-k}\binom{2k}{k} > 0,$$

where $R(t)$ is algebraic at every algebraic $t \in [0, 1]$. If s were algebraic, (5.1) would imply that $D = h(\theta_s - \theta_u) - m(\theta_v - \theta_u)$ is algebraic. Its exponential e^{iD} is also algebraic. The Hermite–Lindemann theorem therefore forces $D = 0$. The same fraction r would then be obtained from both $J_{1,b}$ and $J_{1,1}$, contrary to Lemma 5.3. This proves transcendence for every remaining pair and completes the classification. □

Remark 5.6 (What the argument does and does not use). The rationality of r is used when forming the integer combination of algebraic group points. The general theorem does not replace rational r by arbitrary algebraic r. Nor does it rely on the false assertion that every primitive cyclotomic factor of these Jacobians is simple. The relevant irreducibility is that of a rational homology module under one finite-order automorphism; this is a different property. Finally, the proof treats singular crossings of the planar spiral through its smooth auxiliary curve and a specified monotone radial branch.

5.5 Algebraic-ratio refinements

Although the distinction in Remark 5.6 is substantive, some subfamilies also admit algebraic division ratios. Reduction of differentials gives the following refinements.

Proposition 5.7 (Algebraic-ratio refinements). *Let $0 \leq u < v \leq 1$ be algebraic and let $r \in \overline{\mathbb{Q}} \cap (0, 1)$.*

1. *For $a = 1$ and even b, the solution of (5.1) is algebraic.*
2. *For $a = 1$ and odd $b \geq 3$, that solution is transcendental.*

3. *Suppose $a \geq 2$ and $b = 2a\ell + c$ with $\ell \geq 1$ and $0 < c < a$. Then that solution is transcendental.*

The coprimality convention $(a, b) = 1$ may be retained in these statements.

Proof. The polynomial substitution already proves the first assertion. For the second, (5.6) implies that

$$(\theta_s - \theta_u) - r(\theta_v - \theta_u)$$

is algebraic if s is algebraic. This number is an algebraic linear form in logarithms of algebraic numbers. Such a form is either zero or transcendental by Baker's theorem, so it is zero; see [2, Proposition 15.10 and Remark 15.11]. Lemma 5.3 then gives the same contradiction as before.

For the third assertion, on $y^2 = 1 - x^{2a}$ one has

$$d(x^{b-2a}y) = (b-2a)\frac{x^{b-2a-1}\,dx}{y} - (b-a)\frac{x^{b-1}\,dx}{y}.$$

Iterating this identity gives

$$J_{a,b}(t) = CJ_{a,c}(t) + S(t), \qquad C \in \mathbb{Q}_{>0}, \tag{5.7}$$

where $S(t)$ is algebraic at algebraic t. The differential defining $J_{a,c}$ is holomorphic because $c < a$. If the proposed division point were algebraic, the corresponding algebraic linear combination of its holomorphic integrals would be algebraic, hence zero. The latter fact is the holomorphic case of the analytic subgroup theorem; it also follows from [2, Proposition 13.11], by placing the finitely many paths on a product of Jacobians and absorbing their algebraic coefficients into the invariant differential. Thus the normalized partial integrals with weights b and c would both equal r, contradicting Lemma 5.3. □

5.6 Sinusoidal spirals and Erdős lemniscates

Let $q = a/b > 0$ be reduced. On a chosen radial half-leaf of the sinusoidal spiral

$$r^q = 2\cos(q\theta),$$

the radius increases from 0 to $2^{1/q}$. Direct differentiation of the polar equation gives

$$ds_{\text{arc}} = \frac{dr}{\sqrt{1 - r^{2q}/4}}.$$

On putting $r = 2^{b/a}t^b$, its length from the origin to parameter s is

$$b\,2^{b/a}J_{a,b}(s). \tag{5.8}$$

An endpoint on this branch is algebraic if and only if its radius, equivalently its parameter s, is algebraic. For the nontrivial direction, if r is algebraic then $e^{iq\theta}$ satisfies a quadratic equation over $\overline{\mathbb{Q}}$, and $(e^{i\theta})^a = (e^{iq\theta})^b$ makes $e^{i\theta}$ algebraic.

Corollary 5.8 (All positive rational spiral parameters)**.** *For a positive rational sinusoidal spiral, a nontrivial rational length division of a monotone half-leaf, or of any subarc of that half-leaf with algebraic endpoints, has algebraic coordinates exactly when*

$$q \in \left\{1, \frac{3}{2}, 2, 3\right\} \cup \left\{\frac{1}{2k} : k \geq 1\right\}.$$

For every other positive rational q, the radius of every such division point is transcendental, and the point is not algebraic.

Corollary 5.9 (Division of the integer-petal lemniscates; result 73)**.** *For $n \geq 1$, consider*

$$C_n = \{z \in \mathbb{C} : |z^n - 1| = 1\}.$$

Every internal rational length division point on a monotone half-leaf is algebraic for $n = 1, 2, 3$, and has transcendental radius for every $n \geq 4$. The same assertions hold for subarcs with algebraic endpoints contained in one monotone half-leaf.

For $n \geq 4$, both Cartesian coordinates of every such nonalgebraic division point are separately transcendental. In a cyclic traversal of all petals starting at the origin, the j-th position of an m-fold equal-length division, $0 \leq j < m$, is algebraic exactly when

$$m \mid 2nj.$$

These exceptional positions are the origin and petal tips. There are exactly $\gcd(m, 2n)$ *algebraic indexed positions among the m positions; all m positions are algebraic if and only if $m \mid 2n$.*

Proof. Here $q = n$, and (5.8) has $(a, b) = (n, 1)$. This proves the half-leaf statement. The real equation

$$(X^2 + Y^2)^n - 2\,\mathrm{Re}(X + iY)^n = 0$$

is monic of degree $2n$ as a polynomial in either coordinate, with coefficients polynomial in the other. If one coordinate of a point were algebraic, the other and its radius would also be algebraic.

There are $2n$ equal-length half-leaves in one traversal. Its division position j/m lies at a half-leaf endpoint precisely when $2nj/m$ is an integer. Otherwise it corresponds, after reversing the orientation if necessary, to an internal rational fraction of one half-leaf, to which the first statement applies. The congruence $2nj \equiv 0 \pmod{m}$ has exactly $\gcd(m, 2n)$ solutions modulo m, proving the two counting assertions. □

The final counting statement concerns arclength positions: the same geometric origin is visited between consecutive petals. It is not a claim that all those indexed positions are distinct planar points. The midpoint of a complete leaf is its algebraic tip, which is why "half-leaf" cannot be omitted from the nontrivial-division assertion. Tamiozzo established the cases $n \leq 3$ and asked about $n > 3$ in [1, Remark 5.7]. Corollary 5.9 answers that specific question through the rational-quantile classification above. No assertion of historical priority is made.

5.7 Polynomial images of the circle

An explicit contrast is supplied by

$$z(t) = (1 + e^{it})^m, \qquad -\pi \leq t \leq \pi, \qquad m \in \mathbb{Z}_{>0}.$$

On either monotone radial half of this parametrization, $r = 2^m \cos^m(t/2)$ and $\theta = mt/2$; thus the corresponding spiral parameter is $q = 1/m$. Its total length is

$$L_m = m\, 2^{m+1} J_{1,m}(1).$$

For even m this is rational. For $m = 2k + 1$ it is

$$L_{2k+1} = 2(2k + 1)\binom{2k}{k}\pi.$$

The finite self-intersections do not affect these lengths: each monotone radial half is injective, and any intersections between the two halves are isolated.

Example 5.10 (A parity contrast)**.** For even m, every internal algebraic length ratio on a radial half-arc with algebraic endpoints produces an algebraic point. For odd $m \geq 3$, every such ratio produces a point with transcendental radius, by Proposition 5.7. The first few total lengths are

$$L_1 = 2\pi, \qquad L_2 = 16, \qquad L_3 = 12\pi, \qquad L_4 = \frac{256}{3}, \qquad L_5 = 60\pi.$$

In particular, being an algebraic multiple of π does not ensure algebraicity of equal-arclength division points.

Example 5.11 (The cubic image)**.** For $z = (1 + e^{it})^3$, the radius is $8s^3$. The midpoint of its monotone radial half-arc is specified by the unique $s \in (0, 1)$ with

$$\arcsin s - s\sqrt{1 - s^2} = \frac{\pi}{4}.$$

This s is transcendental, as is $8s^3$. Equivalently, $x = s^2$ is the median of $\mathrm{Beta}(3/2, 1/2)$. The planar image has equation

$$(X^2 + Y^2 - 2X)^3 = 27(X^2 + Y^2)^2.$$

This equation is monic of degree six in either Cartesian coordinate. Consequently both coordinates of the indicated midpoint are transcendental: algebraicity of one would imply algebraicity of the other and of the radius.

These conclusions concern exact arithmetic values and exact division points. They neither obstruct numerical approximation nor imply algebraic independence of the different quantiles.

6 Algebraic levels and torsion packets for generalized lemniscatic integrals

For an integer $n \geq 2$, put

$$I_n(s) = \int_0^s \frac{dt}{\sqrt{1 - t^{2n}}}, \qquad F_n(s) = \frac{I_n(s)}{I_n(1)} \qquad (0 \leq s \leq 1). \tag{6.1}$$

The square root is positive on $(0, 1)$. The endpoint singularity is integrable, and F_n is a continuous strictly increasing bijection of $[0, 1]$ onto itself. A rational division problem asks whether $F_n^{-1}(q)$ is algebraic for $q \in \mathbb{Q}$. In this section we allow *every algebraic real level* q. The new input relative to rational division is a torsion consequence of the analytic subgroup theorem, followed by explicit, previously established classifications of torsion points on curves. The torsion classifications themselves are not new results of this paper.

Theorem 6.1. *Let $n = 4$, or let $n \geq 5$ be an odd prime. Then*

$$s \in \overline{\mathbb{Q}} \cap (0, 1) \quad \Longrightarrow \quad F_n(s) \notin \overline{\mathbb{Q}}.$$

Equivalently, for every $q \in \overline{\mathbb{Q}} \cap (0, 1)$, the unique real solution $s_{n,q} \in (0, 1)$ of

$$\int_0^{s_{n,q}} \frac{dt}{\sqrt{1 - t^{2n}}} = q \int_0^1 \frac{dt}{\sqrt{1 - t^{2n}}} \tag{6.2}$$

is transcendental.

The equivalence in the theorem uses only strict monotonicity. Notice that neither assertion follows from the transcendence of the two integrals separately: a quotient of transcendental numbers can be algebraic. The proof will exclude that possibility through the arithmetic of the associated Abel–Jacobi point.

6.1 A torsion criterion for an algebraic ratio

Let $A/\overline{\mathbb{Q}}$ be an abelian variety and let η be a nonzero invariant algebraic differential on A. We also write η for its linear functional on $\mathrm{Lie}(A)$. Call (A,η) *minimal for the differential* if no positive-dimensional abelian subvariety $B \subset A$ satisfies $\eta|_B = 0$.

This condition can always be obtained by taking a quotient. Indeed, the sum of two abelian subvarieties annihilated by η is again annihilated by η. Finite-dimensionality therefore gives a largest such connected abelian subvariety B. The differential descends to A/B, and the descended pair is minimal. All these objects and maps are defined over $\overline{\mathbb{Q}}$.

We use the following form of the analytic subgroup theorem. If G is a connected commutative algebraic group over $\overline{\mathbb{Q}}$, $z \in \mathrm{Lie}(G)_{\mathbb{C}}$ has algebraic exponential, and an algebraic linear subspace $V \subset \mathrm{Lie}(G)$ contains z after scalar extension, then there is a connected algebraic subgroup $H \subset G$ such that

$$z \in \mathrm{Lie}(H)_{\mathbb{C}} \subset V_{\mathbb{C}}.$$

This follows directly from the annihilator formulation in [2, Theorem 6.2].

Lemma 6.2 (Algebraic ratio to a torsion logarithm)**.** *Suppose (A,η) is minimal for the differential. Let $u, v \in \mathrm{Lie}(A)_{\mathbb{C}}$ satisfy*

$$\exp_A(u) \in A(\overline{\mathbb{Q}}), \qquad \exp_A(v) \in A(\overline{\mathbb{Q}})_{\mathrm{tors}}.$$

If $q \in \overline{\mathbb{Q}}^{\times}$ and $\eta(u) = q\eta(v)$, then $\exp_A(u)$ is a torsion point.

Proof. Apply the analytic subgroup theorem to $G = A \times A$, the logarithm (u, v), and the algebraic hyperplane

$$V_q = \{(U,V) : \eta(U) - q\eta(V) = 0\}.$$

It gives a connected abelian subvariety $H \subset A^2$ containing (u, v) in its complex Lie algebra and having Lie algebra contained in V_q. Let K be the identity component of the kernel of the second projection $\mathrm{pr}_2 : H \to A$. It has the form $B \times \{0\}$ for an abelian subvariety $B \subset A$. The hyperplane condition implies $\eta|_B = 0$. Minimality forces $B = 0$, so $\mathrm{pr}_2|_H$ has finite kernel.

The point $h = (\exp_A(u), \exp_A(v))$ belongs to $H(\mathbb{C})$. Choose an integer $m \geq 1$ killing its second coordinate. Then $[m]h$ lies in the finite kernel of $\mathrm{pr}_2|_H$, hence is torsion. Consequently h is torsion, and so is its first coordinate. □

Remark 6.3. The lemma does not require A to be simple, and it does not require v to be a period: a logarithm of any torsion point suffices. These two features are useful here. In particular, the genus-two Jacobian occurring for $n = 4$ must not be treated as a simple abelian variety. Minimality of the differential is the property actually needed.

6.2 The curve carrying the differential and its minimal quotient

Let C_n be the smooth projective model of

$$v^2 = u(1 - u^n). \tag{6.3}$$

The polynomial on the right is squarefree, and

$$g(C_n) = \left\lfloor \frac{n}{2} \right\rfloor. \tag{6.4}$$

The differential

$$\omega_n = \frac{du}{2v} \tag{6.5}$$

is a nonzero holomorphic differential on C_n. For $0 \le s \le 1$, consider the real path

$$P(s) = \bigl(s^2, s\sqrt{1-s^{2n}}\bigr). \tag{6.6}$$

It runs from $W_0 = (0,0)$ to $W_1 = (1,0)$, and substitution gives

$$\int_{W_0}^{P(s)} \omega_n = I_n(s). \tag{6.7}$$

The integrals are taken along this specified path. Although the displayed affine formula uses branch-point endpoints, the path and the differential are regular in local parameters on the smooth curve.

Both W_0 and W_1 are Weierstrass points. In particular,

$$\operatorname{div}\left(\frac{u-1}{u}\right) = 2W_1 - 2W_0, \qquad [W_1 - W_0] \in J(C_n)[2]. \tag{6.8}$$

Let $j_n : C_n \to J_n = J(C_n)$ be the Abel–Jacobi map based at W_0, and let Ω_n be the invariant differential with $j_n^*\Omega_n = \omega_n$. Denote its minimal quotient by A_n.

Proposition 6.4 (Cyclotomic lower bound)**.** *For every $n \ge 2$,*

$$\dim A_n \ge \frac{\varphi(2n)}{2}.$$

In particular, $A_n = J_n$ when $n = 4$ or when n is an odd prime. Here the equality means that the largest connected abelian subvariety annihilated by Ω_n is zero.

Proof. Choose $\zeta_{2n} = e^{\pi i/n}$ and set $\zeta_n = \zeta_{2n}^2$. The automorphism

$$\rho(u,v) = (\zeta_n u, \zeta_{2n} v)$$

of C_n fixes W_0 and satisfies

$$\rho^*\omega_n = \zeta_{2n}\omega_n. \tag{6.9}$$

It therefore induces an automorphism of J_n with the same eigenvalue on Ω_n. If an abelian subvariety is annihilated by Ω_n, its image under ρ is also annihilated. The largest such subvariety is thus ρ-stable, and ρ descends to A_n.

On A_n, the descended holomorphic differential is still an eigenvector with eigenvalue the primitive $2n$-th root ζ_{2n}. The action on rational first cohomology has dimension $2\dim A_n$ and rational characteristic polynomial. Its complexification contains the holomorphic cohomology and hence the eigenvalue ζ_{2n}. Its characteristic polynomial must therefore be divisible by Φ_{2n}, of degree $\varphi(2n)$. This proves the lower bound.

For $n = 4$, the bound is $\varphi(8)/2 = 2 = g(C_4)$. For an odd prime p, it is $\varphi(2p)/2 = (p-1)/2 = g(C_p)$. Since A_n is a quotient of J_n, equality of dimensions forces the connected kernel to be zero in these cases. □

This argument deliberately avoids a simplicity assertion for $J(C_4)$. The simple cyclotomic factors for odd primes are also described in [1, Proposition 4.8], but their simplicity is not needed in the proof just given.

Corollary 6.5. *Suppose $n = 4$ or $n \ge 5$ is an odd prime. If $s \in \overline{\mathbb{Q}} \cap (0,1)$ and $F_n(s) \in \overline{\mathbb{Q}}$, then $[P(s) - W_0]$ is torsion in J_n.*

Proof. Let u_s and v_1 be the Abel logarithms obtained by integrating from W_0 to $P(s)$ and W_1, respectively, along the chosen path. Their exponentials are $[P(s) - W_0]$ and $[W_1 - W_0]$. The former is algebraic and the latter is torsion by (6.8). Moreover,

$$\Omega_n(u_s) = I_n(s) = F_n(s) I_n(1) = F_n(s)\Omega_n(v_1).$$

The coefficient $F_n(s)$ is positive. Proposition 6.4 makes (J_n, Ω_n) minimal, so Lemma 6.2 applies. □

6.3 Excluding the real interval from the torsion packets

Proof of Theorem 6.1 for $n = 4$. Write $\zeta_8 = e^{\pi i/4}$ and choose $c \in \overline{\mathbb{Q}}$ with $c^2 = \zeta_8$. The transformation

$$X = \zeta_8 u, \qquad Y = cv \tag{6.10}$$

identifies $C_4 : v^2 = u - u^5$ with the smooth projective curve

$$B : Y^2 = X^5 + X. \tag{6.11}$$

Indeed $X^5 + X = \zeta_8(u - u^5) = Y^2$. Let ∞ denote the unique point at infinity on this odd-degree model.

Boxall and Grant classify the entire torsion packet of B based at ∞ in [12, Theorem (0.3)]. It consists of ∞, the five finite Weierstrass points, and sixteen further points. The possible finite X-coordinates are exactly the roots of

$$X^5 + X = 0, \qquad X^4 + 4X^2 + 1 = 0, \qquad X^4 - 4X^2 + 1 = 0. \tag{6.12}$$

The sixteen non-Weierstrass points have order six in this embedding. The reference uses ∞ as base point, whereas our base point maps to $(0, 0)$. This change does not affect the torsion condition: the function X has divisor $2(0, 0) - 2\infty$, so $[(0, 0) - \infty]$ is a 2-torsion point.

At $P(s)$ with $0 < s < 1$, we have $u = s^2 \in (0, 1)$ and $X = \zeta_8 u$. The first equation in (6.12) is impossible because $X \neq 0$ and $X^4 + 1 = 1 - u^4 > 0$. The other two equations become

$$-u^4 + 4iu^2 + 1 = 0, \qquad -u^4 - 4iu^2 + 1 = 0,$$

both impossible because their imaginary parts are nonzero. Thus no point on the open path belongs to this torsion packet. If s and $F_4(s)$ were both algebraic, Corollary 6.5 would give precisely such a torsion point. This proves the assertion. □

Proof of Theorem 6.1 for odd primes $p \geq 5$. The transformation

$$X = -\frac{1}{u}, \qquad Y = \frac{iv}{u^{(p+1)/2}} \tag{6.13}$$

identifies the smooth projective model of C_p with

$$D_p : Y^2 = X^p + 1. \tag{6.14}$$

Indeed both Y^2 and $X^p + 1$ equal $(u^p - 1)/u^p$. The inverse is rational, and the birational map extends to an isomorphism of smooth projective curves. The base point W_0 maps to the unique point ∞ of D_p: u has a double zero at W_0, so X has a double pole there. Hence the base point is exactly the one used in the following torsion classification.

Arul's classification [13, Theorem 6.2.21] gives the whole torsion packet of D_p based at ∞. For $p \geq 7$, its points are precisely ∞ and the points with $X = 0$ or $Y = 0$. For $p = 5$ there are, in addition, the ten points

$$(X, Y) = \left(4^{1/5}\zeta_5^j,\ \pm\sqrt{5}\right), \qquad j = 0, 1, 2, 3, 4. \tag{6.15}$$

The latter case is due to Coleman and is included with attribution in Arul's theorem. These are statements about the complete geometric torsion packet, not just points of small order or points over a fixed number field.

For $P(s)$ with $0 < s < 1$, formula (6.13) reads

$$X = -s^{-2} < -1, \qquad Y = i\frac{\sqrt{1 - s^{2p}}}{s^p} \in i\mathbb{R}_{>0}. \tag{6.16}$$

It is therefore neither at infinity nor a point with $X = 0$ or $Y = 0$. For $p = 5$, its nonzero purely imaginary Y-coordinate also excludes every point in (6.15). Consequently no point on the open path is torsion relative to W_0. Applying Corollary 6.5 proves the theorem. □

6.4 Explicit levels and geometric division

For example, each of the two equations

$$\int_0^a \frac{dt}{\sqrt{1-t^8}} = \frac{1}{\sqrt{2}} \int_0^1 \frac{dt}{\sqrt{1-t^8}}, \tag{6.17}$$

$$\int_0^b \frac{dt}{\sqrt{1-t^{10}}} = \frac{\sqrt{5}-1}{2} \int_0^1 \frac{dt}{\sqrt{1-t^{10}}} \tag{6.18}$$

has a unique solution in $(0,1)$, and both solutions are transcendental. These examples use algebraic irrational levels, so they are not instances of a rational-division conclusion alone.

There is also a direct interpretation on the Erdős lemniscate

$$\Lambda_n = \{z \in \mathbb{C} : |z^n - 1| = 1\}. \tag{6.19}$$

On the half-leaf in the first quadrant joining the origin to $2^{1/n}$, polar coordinates satisfy $r^n = 2\cos(n\theta)$. If $r = 2^{1/n}s$, the arclength from the origin is $2^{1/n}I_n(s)$; hence its fraction of the half-leaf's total arclength is exactly $F_n(s)$.

Corollary 6.6. *Let $n = 4$ or let $n \geq 5$ be an odd prime. For every algebraic $q \in (0,1)$, the point at arclength fraction q on this half-leaf has transcendental radius, transcendental real coordinate, and transcendental imaginary coordinate.*

Proof. Its radius is $2^{1/n}s_{n,q}$ and is transcendental by Theorem 6.1. Write the point as $x + iy$. If either x or y were algebraic, the equation

$$\big((x+iy)^n - 1\big)\big((x-iy)^n - 1\big) = 1$$

would be a nonconstant polynomial equation with algebraic coefficients for the other coordinate. Its leading coefficient in that coordinate is 1. Both coordinates would then be algebraic, and so would $\sqrt{x^2+y^2}$, a contradiction. □

Remark 6.7 (Scope of the classification). The theorem covers $n = 4$ and every odd prime at least five; it does not assert the same conclusion for all composite n. For other n, a torsion conclusion in the minimal differential quotient need not imply torsion in the entire Jacobian, and a torsion-packet theorem for the entire curve cannot simply be applied to that quotient. The proof above isolates and verifies this issue before invoking the packet classifications. Its conclusion is transcendence of each specified quotient or inverse value, not algebraic independence of different values. No claim of historical priority for these corollaries is made here.

Part III
Explicit independent Abelian integrals

7 Independent hyperelliptic integrals

The passage from an algebraic curve to its Jacobian permits a stronger conclusion than the transcendence of a single positive integral. In this section we give explicit families of incomplete integrals whose coordinates are linearly independent over $\overline{\mathbb{Q}}$. There are two distinct constructions. The first uses positivity of a Vandermonde determinant and works in every genus, for at most g ordered intervals on a genus-g curve. The second uses independent quadratic characters and a local formal-group argument; it produces infinitely many independent integrals on one fixed

genus-two curve. The transcendence input is the analytic subgroup theorem. The endomorphism-ring input is an existing theorem of Zarhin. These inputs, rather than a new transcendence theorem for arbitrary periods, account for the conclusions below.

Throughout this section, a choice of logarithm of a nonzero algebraic number means any complex logarithm. Set

$$\mathcal{L}_{\mathrm{alg}} = \overline{\mathbb{Q}} + \mathrm{span}_{\overline{\mathbb{Q}}}\{\lambda \in \mathbb{C} : e^{\lambda} \in \overline{\mathbb{Q}}^{\times}\}. \tag{7.1}$$

In particular, 1 and π belong to this space. Our separation statements concern a vector space of logarithms, not the field generated by logarithms.

7.1 The analytic subgroup input and a separation lemma

We use the following precise form of the analytic subgroup theorem [2, Theorem 6.2]. If $G/\overline{\mathbb{Q}}$ is a connected commutative algebraic group and $u \in \mathrm{Lie}(G)_{\mathbb{C}}$ satisfies $\exp_G(u) \in G(\overline{\mathbb{Q}})$, there is a connected algebraic subgroup $H \subseteq G$ such that

$$u \in \mathrm{Lie}(H)_{\mathbb{C}}, \qquad \mathrm{Ann}_{\overline{\mathbb{Q}}}(u) = \mathrm{Ann}_{\overline{\mathbb{Q}}}(\mathrm{Lie}(H)).$$

Here an annihilator is taken in the algebraic cotangent space of G. Consequently, every algebraic linear form that vanishes at u vanishes on $\mathrm{Lie}(H)$. The inclusion of the particular logarithm u in this Lie algebra is important: one must not replace it merely by a statement about the algebraic point $\exp_G(u)$.

Lemma 7.1 (Logarithms on a power of a simple abelian variety)**.** *Let $A/\overline{\mathbb{Q}}$ have dimension g and satisfy $\mathrm{End}_{\overline{\mathbb{Q}}}(A) = \mathbb{Z}$. Fix an algebraic basis of $\mathrm{Lie}(A)$, and write $u_j = (u_{1j}, \dots, u_{gj})$ for $1 \le j \le r$. Suppose that*

$$\exp_A(u_j) \in A(\overline{\mathbb{Q}}) \quad \textit{and} \quad \sum_{j=1}^{r} m_j u_j = 0,\ m_j \in \mathbb{Z} \implies m_1 = \cdots = m_r = 0.$$

Then all gr coordinates u_{kj} are linearly independent over $\overline{\mathbb{Q}}$. More precisely,

$$\sum_{j=1}^{r}\sum_{k=1}^{g} c_{kj}u_{kj} \in \mathcal{L}_{\mathrm{alg}}, \qquad c_{kj} \in \overline{\mathbb{Q}}, \quad \implies \quad c_{kj} = 0 \textit{ for all } k, j. \tag{7.2}$$

Proof. First, A is absolutely simple. Indeed, a proper nonzero abelian subvariety would, by Poincare reducibility, yield a nontrivial idempotent in $\mathrm{End}^0(A) = \mathbb{Q}$. We also need the following standard description of proper abelian subvarieties of A^r. If $B \subsetneq A^r$ is such a subvariety, then a nonzero homomorphism $A^r \to A$ vanishes on B. To see this, pass to the isogeny category, where the quotient A^r/B is a nonzero power of A, project to one factor, and clear denominators. Since $\mathrm{End}(A) = \mathbb{Z}$, the homomorphism has the form

$$(a_1, \dots, a_r) \longmapsto \sum_j [m_j]a_j$$

for a nonzero integer row $(m_1, \dots, m_r)$. It follows from the hypothesis on the u_j that no proper abelian subvariety of A^r has Lie algebra containing $u = (u_1, \dots, u_r)$.

Suppose now that the left side of (7.2) equals $c + \sum_{h=1}^{s} b_h\lambda_h$, where $c, b_h \in \overline{\mathbb{Q}}$ and $e^{\lambda_h} = \alpha_h \in \overline{\mathbb{Q}}^{\times}$. Consider

$$G = A^r \times \mathbb{G}_a \times \mathbb{G}_m^s, \qquad v = (u, 1, \lambda_1, \dots, \lambda_s).$$

Its exponential is algebraic. The assumed equality supplies an algebraic linear form ℓ with $\ell(v) = 0$. Apply the stated analytic subgroup theorem and let H be the resulting connected subgroup.

At the level of Lie algebras, a connected algebraic subgroup of a product of an abelian variety and a commutative affine algebraic group splits between these two factors. One way to see this is to apply the subgroup version of Goursat's lemma: any relation between the factors would give a common connected quotient, simultaneously proper and affine, hence trivial. Finite kernels do not affect Lie algebras. Thus, if B is the abelian projection of H, then $\mathrm{Lie}(B) \times \{0\} \subseteq \mathrm{Lie}(H)$. As $u \in \mathrm{Lie}(B)_{\mathbb{C}}$, the preceding paragraph forces $B = A^r$. The form ℓ must therefore vanish on the entire A^r direction. In the chosen algebraic basis, this says exactly that all the c_{kj} vanish. □

Remark 7.2. The hypothesis is independence of the logarithm vectors over $\mathbb{Z}$, not algebraic independence of their coordinates. The conclusion is also linear independence, not algebraic independence. For example, a determinant identity involving products of integrals is not decided by this lemma. Separation from $\mathcal{L}_{\mathrm{alg}}$ does imply that the exponential of any nonzero algebraic linear combination of the coordinates is transcendental. If U is such a combination, $e^U \in \overline{\mathbb{Q}}$ would place U in (7.1). Applying this observation to iU also shows that $\sin U$ and $\cos U$ are transcendental: an algebraic value of either would make e^{iU} a root of a quadratic polynomial over $\overline{\mathbb{Q}}$.

7.2 A positive determinant in every genus

For $g \geq 2$, put

$$f_g(x) = x^{2g+1} - x + 1, \qquad C_g : y^2 = f_g(x), \qquad J_g = \mathrm{Jac}(C_g), \tag{7.3}$$

where C_g denotes the smooth projective model. These curves have genus g. On $[0,1]$ one has $f_g(x) = 1 - x + x^{2g+1} > 0$. We always take the positive square root there.

Proposition 7.3 (The endomorphism input)**.** *For every $g \geq 2$,*

$$\mathrm{End}_{\overline{\mathbb{Q}}}(J_g) = \mathbb{Z}.$$

Proof. Zarhin's theorem states that in characteristic zero the Jacobian of $Y^2 = h(X)$ has endomorphism ring $\mathbb{Z}$ when $\deg h \geq 5$ and the Galois group of h is the full symmetric or alternating group. His Example 2.2(1) explicitly applies this to $h(X) = X^n - X - 1$ for every $n \geq 5$ [8, Theorem 2.1 and Example 2.2(1)]. For odd $n = 2g+1$, the change of variables

$$X = -x, \qquad Y = iy$$

identifies C_g over $\overline{\mathbb{Q}}$ with that curve. Endomorphism rings are unchanged by this isomorphism. Thus no unverified assertion about a generic specialization is needed for (7.3). □

Lemma 7.4 (An interval moment determinant)**.** *Let w be positive and continuous on $[0,1]$, and let*

$$0 \leq a_1 < b_1 \leq a_2 < b_2 \leq \cdots \leq a_r < b_r \leq 1.$$

Then

$$\det\left(\int_{a_j}^{b_j} t^{k-1} w(t)\,dt\right)_{1\leq k,j\leq r} = \int_{\prod_j [a_j,b_j]} \prod_{i<j}(t_j - t_i) \prod_{j=1}^{r} w(t_j)\,dt_1 \cdots dt_r > 0. \tag{7.4}$$

Proof. Multilinearity of the determinant and Fubini's theorem give the equality. The determinant of $(t_j^{k-1})_{k,j}$ is the displayed Vandermonde product. Every factor is nonnegative on the integration box, and all factors are strictly positive in its interior. The box has positive measure and w is positive, proving the strict inequality. Common endpoints between adjacent intervals have measure zero and cause no problem. □

Theorem 7.5 (Ordered interval integrals; result 72)**.** *Let $g \geq 2$, let $1 \leq r \leq g$, and let the endpoints in Lemma 7.4 be real algebraic numbers. Define*

$$M_{k,j} = \int_{a_j}^{b_j} \frac{t^k\,dt}{\sqrt{t^{2g+1} - t + 1}}, \qquad 0 \leq k \leq g-1, \quad 1 \leq j \leq r. \tag{7.5}$$

All gr numbers $M_{k,j}$ are linearly independent over $\overline{\mathbb{Q}}$, even modulo $\mathcal{L}_{\mathrm{alg}}$ in the sense of (7.2). In particular, the family consisting of 1, π, and these gr integrals is linearly independent over $\overline{\mathbb{Q}}$.

Proof. The forms

$$\omega_k = x^k\,dx/y, \qquad 0 \leq k \leq g-1,$$

are an algebraic basis of the regular differentials on C_g. Via the Abel–Jacobi map they give an algebraic cotangent basis on J_g. For each j, lift the real interval $[a_j, b_j]$ to the positive branch of C_g. The vector

$$u_j = (M_{0,j}, \ldots, M_{g-1,j})$$

is a logarithm on J_g of the algebraic divisor class of the difference of its two endpoints. Lemma 7.4, applied to $w(t) = f_g(t)^{-1/2}$ and the first r coordinates, shows that the u_j are linearly independent even over $\mathbb{R}$. They are therefore independent over $\mathbb{Z}$. Proposition 7.3 and Lemma 7.1 prove separation from $\mathcal{L}_{\mathrm{alg}}$.

For the final assertion, a relation involving 1, π and the $M_{k,j}$ first has all integral coefficients equal to zero by separation. The remaining relation between 1 and π is trivial by the transcendence of π. □

Remark 7.6. The restriction $r \leq g$ is used exactly at the positive determinant step. It is a sufficient condition for this proof, not a claim that more than g intervals must yield dependence. The construction below supplies infinitely many independent endpoint integrals on C_2 by a different argument. In Theorem 7.5, taking $r = g$ gives g^2 explicit independent integrals from g ordered intervals on one curve.

7.3 Four independent integrals and normalized cut probabilities

Specialize to

$$f(x) = x^5 - x + 1, \qquad I_k(x) = \int_0^x \frac{t^k\,dt}{\sqrt{f(t)}} \quad (k = 0, 1).$$

Theorem 7.7 (Every algebraic interior cut; result 71)**.** *For every real algebraic x with $0 < x < 1$, the four numbers*

$$I_0(x), \quad I_1(x), \quad I_0(1), \quad I_1(1) \tag{7.6}$$

are linearly independent over $\overline{\mathbb{Q}}$, even modulo $\mathcal{L}_{\mathrm{alg}}$. Consequently,

$$F(x) = \frac{I_0(x)}{I_0(1)} \quad \text{and} \quad \frac{I_1(x)}{I_1(1)} \tag{7.7}$$

are transcendental. In fact, the ratio of any two distinct members of (7.6) is transcendental.

Proof. Apply Theorem 7.5 to the two intervals $[0, x]$ and $[x, 1]$. Its four independent integrals are

$$I_0(x),\ I_1(x),\ I_0(1) - I_0(x),\ I_1(1) - I_1(x).$$

The rational change of coordinates that replaces the last two entries by their sums with the first two is invertible. It preserves independence and separation from $\mathcal{L}_{\mathrm{alg}}$, proving the first assertion. All four numbers are positive. If the ratio of two distinct entries were $c \in \overline{\mathbb{Q}}$, their difference after multiplication by c would be a nontrivial algebraic linear relation, a contradiction. □

There is an elementary way to see the real-vector hypothesis underlying this argument. The ratio

$$\frac{I_1(x)}{I_0(x)}$$

is the mean of t on $[0,x]$ for the strictly positive weight $f(t)^{-1/2}$. The corresponding mean on $[x,1]$ is strictly larger. Their two moment vectors are therefore nonproportional. This elementary positivity statement does not itself prove transcendence; the arithmetic conclusion follows only after the endomorphism theorem and the analytic subgroup theorem have been applied.

Corollary 7.8 (Affine weights and quantiles)**.** *Let $W(t) = a + bt \in (\overline{\mathbb{Q}} \cap \mathbb{R})[t]$ be nonnegative on $[0,1]$ and not identically zero. Set*

$$Z_W = \int_0^1 \frac{W(t)\,dt}{\sqrt{f(t)}}, \qquad F_W(x) = \frac{1}{Z_W}\int_0^x \frac{W(t)\,dt}{\sqrt{f(t)}}.$$

Then $F_W(x)$ is transcendental for every algebraic $x \in (0,1)$. For every algebraic $q \in (0,1)$, the unique solution $x \in (0,1)$ of $F_W(x) = q$ is transcendental.

Proof. Positivity gives $Z_W > 0$. If $F_W(x) = c \in \overline{\mathbb{Q}}$ at an algebraic interior point, then

$$aI_0(x) + bI_1(x) - caI_0(1) - cbI_1(1) = 0.$$

The coefficients are not all zero because $(a,b) \neq (0,0)$, contrary to Theorem 7.7. The distribution function is continuous and strictly increasing: the nonzero affine polynomial W cannot vanish on an interval. Thus each interior quantile exists uniquely. If it were algebraic, the preceding assertion would contradict the algebraicity of q. □

The same argument works on C_g for any nonzero polynomial weight of degree at most $g-1$ which is nonnegative on $[0,1]$: use the $2g$ independent integrals supplied by the intervals $[0,x]$ and $[x,1]$. This degree restriction keeps the differential in the explicitly used space of regular differentials. No assertion here extends this proof automatically to arbitrary polynomial weights of larger degree.

7.4 An infinite family on the fixed genus-two curve

We next remove the finite-interval limitation by constructing infinitely many independent quadratic characters. The curve, its Jacobian, the positive real branches, and the functions I_0, I_1 remain fixed. For a prime p, put

$$N_p = p^5 - p^4 + 1, \qquad P_p = \left(\frac{1}{p}, \sqrt{\frac{N_p}{p^5}}\right), \qquad D_p = [P_p - \infty] \in J_2(\overline{\mathbb{Q}}). \tag{7.8}$$

The square root in P_p is positive in the real embedding.

Define a strictly increasing sequence of primes recursively by

$$p_1 = 5, \qquad p_n = \min\left\{ p > p_{n-1} : \ p \text{ is prime}, \quad p \nmid 2869 \prod_{j<n} N_{p_j} \right\} \quad (n \geq 2). \tag{7.9}$$

At each stage only finitely many primes are excluded, so the sequence is well defined and infinite. Its initial terms, obtained by integer divisibility alone, are

$$5, 7, 11, 13, 17, 23, 29, 31, 37, 47, 53, 59, 67, 71, 73, \ldots.$$

The sequence is a precise recursive definition; no assumption about prime values of a polynomial is being made.

Lemma 7.9 (A local certificate of infinite order)**.** *If $p \geq 5$ is prime and $p \nmid 2869$, then the point D_p in (7.8) has infinite order.*

Proof. The polynomial discriminant is

$$\operatorname{disc}(x^5 - x + 1) = 2869 = 19 \cdot 151.$$

Since p is odd and does not divide this discriminant, the smooth projective hyperelliptic model has good reduction at p, as does its Jacobian. Work over

$$K_p = \mathbb{Q}_p(\sqrt{pN_p}).$$

Because $N_p \equiv 1 \pmod{p}$, the polynomial $T^2 - pN_p$ is Eisenstein at p. Thus $K_p/\mathbb{Q}_p$ is a ramified quadratic extension with absolute ramification index $e = 2$.

Near the unique point at infinity, the coordinates

$$u = 1/x, \qquad v = y/x^3$$

give the integral chart

$$v^2 = u - u^5 + u^6. \tag{7.10}$$

It is smooth at $(u, v) = (0, 0)$, since the derivative of its defining equation with respect to u is a unit there. At P_p the two coordinates are $u = p$ and $v = \sqrt{pN_p}$, so P_p reduces to ∞. The Abel map therefore places D_p in the kernel of reduction on $J_2(K_p)$. This kernel is the group of maximal-ideal points of the formal group of the smooth Jacobian model. Also $D_p \neq 0$: the Abel map of a positive-genus curve based at one point is injective. Equivalently, a principal divisor $P_p - \infty$ would give a rational function of degree one and make the curve rational.

For completeness, the relevant formal-group fact is that a commutative formal group over the integers of a p-adic field has no nonzero torsion in its maximal-ideal points when $e < p - 1$ [14, Proposition 9 and the following paragraph]. Here is the valuation argument in the needed general dimension. For formal coordinates T one has

$$[p](T) = p(T + \phi(T)) + \psi(T), \qquad \deg \phi \geq 2, \quad \deg \psi \geq p.$$

Normalize the valuation by $v(K_p^\times) = \mathbb{Z}$, so $v(p) = e$. For a nonzero maximal-ideal tuple T, let $s \geq 1$ be the minimum valuation of its coordinates. In a coordinate realizing that minimum, the term pT has valuation $e + s$, whereas the other terms have valuation at least $e + 2s$ or ps. Since $e < p - 1$, $e + s < ps$, so cancellation is impossible and $[p](T) \neq 0$. Multiplication by an integer prime to p is an invertible formal power series, which excludes prime-to-p torsion as well. Any nonzero finite-order point would then produce a nonzero point of prime order, an impossibility. In our case $e = 2 < p - 1$, proving the lemma. □

Lemma 7.10 (Independent sign changes)**.** *For every $r \geq 1$, the classes*

$$[f(1/p_1)], \ldots, [f(1/p_r)] \in \mathbb{Q}^\times/\mathbb{Q}^{\times 2}$$

are independent over $\mathbb{F}_2$. Consequently the compositum of the r quadratic point fields has Galois group $(\mathbb{Z}/2\mathbb{Z})^r$. For each j there is an automorphism σ_j that changes the sign of the ordinate of P_{p_j} and fixes every P_{p_i} with $i \neq j$. On the Jacobian,

$$\sigma_j(D_{p_j}) = -D_{p_j}, \qquad \sigma_j(D_{p_i}) = D_{p_i} \quad (i \neq j).$$

Proof. At the new prime p_j, the valuation of $f(1/p_j) = N_{p_j}/p_j^5$ is -5. Its valuation on every earlier $f(1/p_i)$ is zero by (7.9). In a proposed nonempty product of these square classes, take the largest index with nonzero exponent. The valuation at its corresponding prime is odd, so the

product cannot be a square. This proves independence and the description of the multiquadratic Galois group.

The hyperelliptic involution sends (x, y) to $(x, -y)$ and fixes ∞. The divisor of $x - x(P)$ is $P + \iota(P) - 2\infty$. Consequently it induces multiplication by -1 on the Abel image based at infinity. This proves the displayed action on the D_{p_i}. □

Theorem 7.11 (An infinite family on one curve; result 74)**.** *For the sequence* (7.9)*, the infinite family*

$$I_0(1),\ I_1(1),\quad I_0(1/p_n),\ I_1(1/p_n)\quad (n \geq 1) \tag{7.11}$$

is linearly independent over $\overline{\mathbb{Q}}$*, even modulo* $\mathcal{L}_{\mathrm{alg}}$*. Thus adjoining* 1 *and* π *still leaves a linearly independent family. Every assertion about this infinite family means the corresponding assertion for each finite subfamily.*

Proof. Let $P_0 = (0, 1)$ and $P_* = (1, 1)$, and use the basis $dx/y, x\,dx/y$ of regular differentials. Write

$$u_* = (I_0(1), I_1(1)), \qquad u_n = (I_0(1/p_n), I_1(1/p_n)).$$

These are logarithms of the algebraic Jacobian points

$$Q_* = [P_* - P_0], \qquad Q_n = [P_{p_n} - P_0].$$

We verify exactly the logarithm-vector hypothesis of Lemma 7.1. Suppose that for some r,

$$m_* u_* + \sum_{n=1}^{r} m_n u_n = 0, \qquad m_*, m_n \in \mathbb{Z}. \tag{7.12}$$

Exponentiation gives $[m_*]Q_* + \sum_n [m_n]Q_n = 0$ in $J_2(\overline{\mathbb{Q}})$. The points P_0, P_* are rational. Put $D_0 = [P_0 - \infty]$; then $Q_n = D_{p_n} - D_0$. Applying $\sigma_j - 1$ from Lemma 7.10 to the point relation gives

$$[-2m_j]D_{p_j} = 0.$$

Lemma 7.9 implies $m_j = 0$ for every j. Returning to the exact vector relation (7.12) leaves $m_* u_* = 0$. Its first coordinate is strictly positive, so $m_* = 0$. This last step does not require any assertion about whether Q_* is a torsion point: an exact relation of logarithm vectors is stronger than the exponentiated relation.

All these logarithm vectors are therefore independent over $\mathbb{Z}$. Since $\mathrm{End}(J_2) = \mathbb{Z}$, Lemma 7.1 proves the theorem. The assertion with 1 and π follows as in Theorem 7.5. □

Corollary 7.12 (Normalized values and pairwise ratios)**.** *The family*

$$1,\quad \frac{I_1(1)}{I_0(1)},\quad \frac{I_0(1/p_n)}{I_0(1)},\quad \frac{I_1(1/p_n)}{I_0(1)}\quad (n \geq 1)$$

is linearly independent over $\overline{\mathbb{Q}}$*. The ratio of any two distinct members of* (7.11) *is transcendental. Every nonzero finite algebraic linear combination* U *of the unnormalized members of* (7.11) *is transcendental, and so are* e^U*,* $\sin U$ *and* $\cos U$*.*

Proof. For the first assertion, multiply a putative relation by the nonzero number $I_0(1)$ and apply Theorem 7.11. An algebraic ratio between two distinct entries would likewise give a nontrivial linear relation. The remaining assertions follow from separation from $\mathcal{L}_{\mathrm{alg}}$ and Remark 7.2. □

Remark 7.13 (Scope and provenance). The endomorphism theorem, the analytic subgroup theorem and the formal-group torsion bound are classical inputs, explicitly cited above. The displayed integral families and their independence statements are deductions constructed from

those inputs. No priority claim is made for their particular presentation. Neither the finite determinant argument nor the infinite quadratic-character construction establishes algebraic independence of the integrals, of their normalized values, or of their exponentials. Also, the countably infinite independent family on one fixed curve is not obtained by taking a limit in the $r \le g$ theorem: its proof supplies a separate arithmetic certificate for every finite set of endpoints.

8 Combining the two infinite families

The two independence constructions can be placed in one statement. They use abelian varieties of different dimensions, so no accidental identification of their simple factors is possible.

Corollary 8.1. *Let ℓ_p be the Cassini perimeters in (3.1), let p_n be the sequence (7.9), and let I_0, I_1 denote the integrals on $y^2 = x^5 - x + 1$ in Section 7. Then*

$$\{1, \pi\} \cup \{\ell_p : p \text{ an odd prime}\} \cup \{I_0(1), I_1(1)\} \cup \{I_0(1/p_n), I_1(1/p_n) : n \ge 1\}$$

is linearly independent over $\overline{\mathbb{Q}}$.

Proof. Consider any finite subset. Its abelian logarithms lie in a product of pairwise nonisogenous elliptic curves E_p and a power of the absolutely simple abelian surface J_2. There are no nonzero homomorphisms between J_2 and any E_p: the image of such a homomorphism, or its dual, would contradict simplicity and the different dimensions. In the isogeny category every connected abelian subgroup therefore splits between these isotypic factors. The nonzero elliptic period vectors force all elliptic factors to occur, as in Theorem 3.2. The exact logarithm-vector independence in the proof of Theorem 7.11 forces the whole power of J_2 to occur. Applying the analytic subgroup theorem to a proposed linear relation, with $\mathbb{G}_a$ and $\mathbb{G}_m$ appended for 1 and $2\pi i$, forces the relation to vanish on every factor. All its coefficients are zero. □

This combined statement still concerns linear independence. It gives no algebraic-independence conclusion for products, powers, or arbitrary polynomial expressions in the displayed numbers.

Part IV
Transfers to other mathematical structures

9 Arithmetic transfer principles and applications

This section records the transfers numbered 1–60 in the discussion following [15]. Several are consequences of the same arithmetic input; we therefore prove the common principles before giving the numbered catalogue. Reusing a theorem in another subject does not constitute a new transcendence theorem. The statements below distinguish algebraic independence, transcendence of a single value, and effective decisions. Throughout, algebraic means algebraic over $\mathbb{Q}$.

9.1 Exponential integrals and a measure with independent moments

Put

$$\mathrm{Ein}(z) = \int_0^z \frac{1 - e^{-t}}{t}\, dt, \qquad \mathcal{E} = \overline{\mathbb{Q}}(e^{\alpha} : \alpha \in \overline{\mathbb{Q}}).$$

The next argument strengthens the moment input by retaining arbitrary algebraic exponential constants. Its ingredients are the Kolchin–Ostrowski theorem for antiderivatives and the Siegel–Shidlovskii theorem for E-functions [16, 17].

Theorem 9.1. *For distinct nonzero $a_1, \dots, a_m \in \overline{\mathbb{Q}}$, the numbers $\mathrm{Ein}(a_1), \dots, \mathrm{Ein}(a_m)$ are algebraically independent over $\mathcal{E}$.*

Proof. It suffices to include finitely many extra exponential constants. Choose algebraic numbers $\beta_1, \dots, \beta_r$ linearly independent over $\mathbb{Q}$ such that every a_j and every exponent occurring in these constants is an integer combination of the β_i. The functions $t_i = e^{\beta_i z}$ are algebraically independent over $\mathbb{C}(z)$: a polynomial relation would be a relation between exponentials with distinct exponents and rational-function coefficients. Let $K = \mathbb{C}(z, t_1, \dots, t_r)$, with derivation $D = \partial_z + \sum_i \beta_i t_i \partial_{t_i}$. Its constant field is $\mathbb{C}$ in its meromorphic realization.

If the primitives $y_j = \mathrm{Ein}(a_j z)$ were algebraically dependent over K, the antiderivative theorem would give

$$DR = \frac{G(t_1, \dots, t_r)}{z}, \qquad R \in K, \quad G = \sum_j c_j (1 - e^{-a_j z}), \quad (c_j) \neq 0,$$

where G is viewed as a Laurent polynomial in the t_i. Distinct nonzero a_j imply $G \neq 0$. Expand R at $z = 0$ over $\mathbb{C}(t_1, \dots, t_r)$. If R has pole order $d \geq 1$, then DR has pole order $d + 1$, because the ∂_z term has strictly larger pole order than the other terms. If R is regular, DR is regular. A nonzero simple pole is impossible in either case. Thus the y_j are algebraically independent over K.

The functions 1, y_j, $e^{-a_j z}$, and $e^{\beta_i z}$ form a homogeneous linear differential system over $\overline{\mathbb{Q}}(z)$ with no singularity away from 0. They are E-functions. Siegel–Shidlovskii at $z = 1$ therefore preserves the transcendence degree $r + m$. The exponentials e^{-a_j} and the specified extra exponentials are Laurent monomials in the e^{β_i}. This proves the assertion for every finite subset of $\mathcal{E}$, and hence for $\mathcal{E}$ itself. □

Define a finite positive measure on $[e^{-1}, 1]$ by

$$d\mu(y) = \frac{1 - y}{-y \log y}\, dy, \qquad m_n = \int y^n \, d\mu(y) = \mathrm{Ein}(n + 1) - \mathrm{Ein}(n), \qquad \mathrm{Ein}(0) = 0. \tag{9.1}$$

Consequently every finite sequence $m_0, \dots, m_N$ is algebraically independent over $\mathcal{E}$. In particular, adjoining e to any such sequence preserves algebraic independence over $\overline{\mathbb{Q}}$. A useful consequence, used repeatedly below, is

$$R \in \overline{\mathbb{Q}}(m_0, \dots, m_N) \setminus \overline{\mathbb{Q}}, \ R > 0 \quad \Longrightarrow \quad \log R \text{ is transcendental.} \tag{9.2}$$

Indeed, a nonconstant rational function of independent variables cannot belong to $\mathcal{E}$; if $\log R$ were algebraic, R would belong to $\mathcal{E}$. This includes $\log \mathrm{Ein}(1)$. General results about logarithms of E-function values are available in [18]; no priority claim is attached to this example.

Proposition 9.2 (Quadrature coordinates)**.** *For a fixed $q \geq 1$, let $x_1 < \cdots < x_q$ and $w_1, \dots, w_q > 0$ be the Gaussian quadrature nodes and weights for μ. All $2q$ numbers x_j, w_j are algebraically independent over $\mathcal{E}$. For $v_j = w_j / m_0$, the $2q$ numbers*

$$m_0, x_1, \dots, x_q, v_1, \dots, v_{q-1}$$

are likewise algebraically independent; the only polynomial relation among the full normalized list is generated by $\sum_j v_j = 1$. The following further families are algebraically independent over $\mathcal{E}$: all nonleading coefficients of one fixed monic orthogonal polynomial; any initial finite list of positive Hankel determinants; and the mass together with the corresponding finite Jacobi recurrence parameters.

Proof. Gaussian exactness gives $m_k = \sum_j w_j x_j^k$ for $0 \le k < 2q$. Thus $2q$ generators contain a field of transcendence degree $2q$. The normalized assertion is a rational change of coordinates. The nonleading coefficients of a degree-q monic polynomial generate a field over which its q roots are algebraic, proving the coefficient assertion. Writing $\Delta_r = \det(m_{i+j})_{0\le i,j<r}$ and $\Delta_0 = 1$, expansion in m_{2r-2} gives $\Delta_r = \Delta_{r-1} m_{2r-2} +$ earlier terms. This triangular change proves independence of the Δ_r. Finally, the usual finite Jacobi recurrence parameters and m_0 rationally recover $m_0, \dots, m_{2q-1}$ and conversely. The poles and residues of the corresponding Stieltjes Padé approximant are precisely the Gaussian nodes and weights. □

Proposition 9.3 (Errors and extremal energies)**.** *The q-node Gaussian error of a polynomial $P \in \overline{\mathbb{Q}}[y]$ is zero if $\deg P \le 2q-1$, and is transcendental if $\deg P \ge 2q$. For fixed real algebraic s, the Christoffel extremal energies*

$$\lambda_n(s) = \min\left\{ \int P^2 \, d\mu : P \in \mathbb{R}[y],\ \deg P \le n,\ P(s) = 1 \right\} \quad (n \ge 0)$$

are jointly algebraically independent, in every finite subfamily, over $\mathcal{E}$. For $q \ge 2$ and integers $r \ge 2$, the Rényi entropy $(1-r)^{-1} \log \sum_j v_j^r$ is transcendental.

Proof. For $M \ge 2q$, the nodes, weights, and errors of $y^{2q}, \dots, y^M$ together generate a field containing all $m_0, \dots, m_M$. The number of generators is $M+1$, so these generators are algebraically independent. Every nontrivial algebraic linear combination of the error coordinates is transcendental. For the second assertion, let P_n be monic orthogonal polynomials and $h_n = \int P_n^2 \, d\mu$. Then

$$\lambda_n(s)^{-1} - \lambda_{n-1}(s)^{-1} = P_n(s)^2/h_n, \qquad h_n = m_{2n} + \text{earlier terms}.$$

Here $P_n(s) \ne 0$, since its roots are transcendental. Together with the odd moments, the $\lambda_n(s)$ therefore form a birational coordinate system on the finite moment field. For entropy, $v_1, \dots, v_{q-1}$ are independent over $\mathcal{E}$, with $v_q = 1 - \sum_{j<q} v_j$. The positive nonconstant polynomial $\sum_j v_j^r$ therefore cannot belong to $\mathcal{E}$, so its logarithm is transcendental by the argument of (9.2). □

Proposition 9.4 (Polynomial ensembles and cumulants)**.** *For the probability measure*

$$\frac{1}{n!\Delta_n} \prod_{i<j} (y_i - y_j)^2 \prod_{i=1}^{n} d\mu(y_i),$$

the partition functions $Z_n = n!\Delta_n$, $n \ge 1$, are jointly algebraically independent over $\mathcal{E}$, and each $\log Z_n$ is transcendental. Every nonconstant symmetric polynomial with real algebraic coefficients has transcendental expectation and transcendental variance. For the normalized measure $\nu = \mu/m_0$, all its cumulants κ_j, $j \ge 1$, together with m_0, are jointly algebraically independent over $\mathcal{E}$.

Proof. Expansion of the Vandermonde determinant expresses each expectation as a rational function of finitely many moments. This rational function is nonconstant for a nonconstant symmetric polynomial S: specialize the measure to exactly n distinct atoms, so the ensemble is supported on permutations of those atoms and the expectation equals the freely varying value of S. The variance is zero at such a specialization, but is strictly positive for the continuous measure (9.1); it too is a nonconstant rational function. Independence of the moments and (9.2) finish these assertions. Normalized moments and cumulants are related triangularly by $\kappa_j = m_j/m_0 +$ a polynomial in earlier normalized moments. □

Corollary 9.5 (Differential equations and finite quantum dynamics)**.** *Let $M(t) = \int e^{ty} \, d\mu(y)$ and $a_0(t) = M'(t)/M(t)$. Neither function satisfies a nonzero algebraic differential equation over $\overline{\mathbb{Q}}(e)(t)$. If H is the $q \times q$ real symmetric Jacobi matrix from Proposition 9.2, $q \ge 3$, its first basis vector has no exact projective revival at any positive time under e^{-itH}.*

Proof. The derivatives $M^{(j)}(0) = m_j$ are algebraically independent over $\overline{\mathbb{Q}}(e)$, as are $a_0^{(j)}(0) = \kappa_{j+1}$. Clear the denominators of a hypothetical differential equation and remove the largest common power of t from its coefficients. Evaluation at 0 would give a nonzero algebraic relation among these jets. This statement concerns the specified coefficient field; it asserts nothing about differential algebraicity over $\mathbb{C}(t)$. For a revival, positivity of all boundary spectral weights requires $T(x_j - x_1) \in 2\pi\mathbb{Z}$ for every j. Three distinct eigenvalues would therefore have a rational gap ratio, contradicting their algebraic independence. This does not exclude approximate revivals. □

9.2 Polynomial beta tilts and exact cut obstructions

Fix distinct primes p, r with $\max(p, r) \geq 5$, and

$$a = i/p, \quad b = j/r, \quad 0 < i < p, \quad 0 < j < r. \tag{9.3}$$

Let $w_0(x) = x^{a-1}(1-x)^{b-1}$, $B = B(a, b)$, and $U(s) = B_s(a, b)$. We use the baseline theorem that

$$1, B, U(s) \text{ are linearly independent over } \overline{\mathbb{Q}} \quad (s \in \overline{\mathbb{Q}} \cap (0, 1)). \tag{9.4}$$

This is the arithmetic input; the applications below do not replace it with an assertion for arbitrary rational beta parameters. Let $W \in (\overline{\mathbb{Q}} \cap \mathbb{R})[x]$ be nonzero and nonnegative on $[0, 1]$. Set $c_W = B^{-1} \int_0^1 W w_0$ and $f_W = W w_0/(c_W B)$. Thus c_W is positive algebraic.

Lemma 9.6 (Polynomial reduction)**.** *For every $R \in \overline{\mathbb{Q}}[x]$ there are unique $c_R \in \overline{\mathbb{Q}}$ and $q_R \in \overline{\mathbb{Q}}[x]$ such that*

$$R = c_R + x(1-x)q_R' + [a - (a+b)x]q_R, \ \int_0^s R w_0 \, dx \quad = c_R U(s) + s^a(1-s)^b q_R(s). \tag{9.5}$$

At algebraic $s \in (0, 1)$ the normalized partial integral $\int_0^s R w_0/(c_W B)$ is algebraic if and only if $c_R = q_R(s) = 0$; its value is then zero. All full polynomial moments of f_W are algebraic, and are rational when W has rational coefficients.

Proof. The displayed differential operator sends x^k to $(k+a)x^k - (k+a+b)x^{k+1}$. Its nonzero leading coefficient gives existence and uniqueness by elimination of descending degrees, leaving a constant. Integration uses $(x^a(1-x)^b)' = [a - (a+b)x]w_0$. The criterion follows immediately from (9.4), and the full moments follow from $B(a+k, b)/B(a, b) = (a)_k/(a+b)_k$. □

Proposition 9.7 (Quantiles and conditional statistics)**.** *For X with density f_W, every interior quantile at an algebraic probability is transcendental. At every algebraic $s \in (0, 1)$, $F(s)$, $1 - F(s)$, and $F(s)/(1 - F(s))$ are transcendental. If $W(s) > 0$, so are the density, hazard, and reverse hazard. For each integer $k \geq 1$, both conditional moments $\mathbb{E}[X^k \mid X \leq s]$ and $\mathbb{E}[X^k \mid X \geq s]$ are transcendental. More precisely, each lower conditional moment generates $\overline{\mathbb{Q}}(U(s))$, and each upper conditional moment generates $\overline{\mathbb{Q}}(B - U(s))$.*

Proof. Apply Lemma 9.6 to W; its partial integral is strictly positive. The density is a nonzero algebraic multiple of $1/B$. The hazard denominators are algebraic combinations of $B, U(s), 1$; (9.4) excludes an algebraic quotient. If the lower conditional moment were algebraic r, applying (9.4) to the integral of $(x^k - r)W$ would force $r = \mathbb{E}[X^k]$. This contradicts strict monotonicity of x^k and positive mass on both sides of s. The upper case is identical. Formula (9.5) expresses each lower conditional moment as a nonconstant fractional-linear function of $U(s)$, and each upper one as such a function of $B - U(s)$; inverting it gives the asserted exact fields. Residual lifetimes follow by subtracting the algebraic cut from the conditional first moments. At a zero of W, the density and hazards may be zero; the condition $W(s) > 0$ is essential. □

Polynomial likelihoods, including arbitrary Bernoulli updates, preserve this family. Aggregated Dirichlet margins have the same conclusion whenever their two resulting parameters meet (9.3), up to nonnegative integer shifts. An order statistic or an iid coherent reliability system has distribution $H(F(x))$ for a nonconstant polynomial H with integer coefficients. At an algebraic level, an algebraic value of $H(F(x))$ would imply an algebraic value of $F(x)$; hence their interior quantiles are transcendental. This also applies to equal-mass grids and monotone transport from a source distribution whose relevant cut probabilities are algebraic. In contrast, finite Gaussian quadrature for f_W has algebraic nodes and weights because all its full moments are algebraic. Its finite atomic distribution cannot reproduce $F(s)$ exactly at an algebraic interior cut.

Theorem 9.8 (Optimization and multivariate polynomial tilts)**.** *Let $p_0 \geq 1$ be an integer and $\tau \in \overline{\mathbb{Q}} \cap (0,1)$. The unique minimizer $s \in (0,1)$ of*

$$\mathbb{E}\{\tau(X-s)_+^{p_0} + (1-\tau)(s-X)_+^{p_0}\}$$

is algebraic if and only if p_0 is even and $\tau = 1/2$. For algebraic $s \in (0,1)$, $W_{p_0}(\mathcal{L}(X), \delta_s)$ is algebraic for even p_0 and transcendental for odd p_0. A Lloyd update from distinct algebraic centers in $(0,1)$ gives transcendental first and last updated centers.

Moreover, tilt independent copies of a rational-parameter beta law satisfying (9.3) by a nonzero nonnegative polynomial $Q \in \mathbb{Q}[x_1, \ldots, x_d]$. All polynomial joint moments remain rational, while every marginal cumulative probability at an algebraic interior cut is transcendental. This applies to even-integer-β Jacobi ensembles and their expected partial power sums.

Proof. Put $m = p_0 - 1$, $A = \mathbb{E}(X-s)_+^m$, and $L = \mathbb{E}(s-X)_+^m$. For $m = 0$ these expressions mean the corresponding side probabilities. The optimum satisfies $\tau A = (1-\tau)L$. If s is algebraic, A, L are positive transcendental partial polynomial integrals, whereas $M = \mathbb{E}(X-s)^m = A + (-1)^m L$ is algebraic. Thus

$$[1 - \tau + \tau(-1)^m]L = \tau M.$$

The coefficient vanishes exactly when p_0 is even and $\tau = 1/2$. In that case the optimum solves the algebraic polynomial equation $\mathbb{E}(X-s)^{p_0-1} = 0$. Strict convexity, or strict monotonicity of the cumulative distribution when $p_0 = 1$, gives uniqueness. The Wasserstein assertion follows by splitting $|X-s|^{p_0}$ at s: for odd p_0 it is a full polynomial moment plus twice a nonzero partial polynomial integral. Algebraicity of its positive p_0th root is equivalent to algebraicity of the integral. The two outer Lloyd cells are truncated intervals with algebraic cuts, so Proposition 9.7 applies. In particular no entirely algebraic stationary codebook with at least two centers is possible; with at least four centers, the two disjoint outer pairs force at least two transcendental centers.

Expanding Q shows that its normalizer and all joint polynomial moments are rational. Integrating out all but one variable leaves a nonzero nonnegative polynomial tilt, to which the preceding proposition applies. For an even-integer-β Jacobi ensemble, the Vandermonde factor is such a polynomial. Symmetry reduces the expected partial sum $\mathbb{E}\sum_i \lambda_i^m \mathbf{1}_{\lambda_i \leq s}$, $m \geq 0$, to a positive marginal partial polynomial integral. The conclusion does not assert that every individual counting probability is transcendental. □

9.3 Symbolic reduction, derivatives, and exact number fields

For clarity we record the scope of the DBKZ input from [15]. Let $a, b, d \in \frac{1}{5}\mathbb{Z}$, $a, b > -1$, and $c > 0$ algebraic. Assume that $(5a, 5b, -5d, 5(d-a-b)) \bmod 5$ is a permutation of $(h, h, h, 2h)$, $h \in \{1, 2, 3, 4\}$, and that c has a good reduction with value 25 modulo 31, or value 7 modulo 11, as in the baseline criterion. With

$$g = x^a(1-x)^b(1+cx)^{-d}, \qquad D = \int_0^1 g\,dx, \qquad M = \frac{1}{D}\int_0^1 xg\,dx,$$

that criterion gives M transcendental and the exact reduction

$$R = u + vx + \mathcal{L}S, \quad \mathcal{L}S = S' + \left(\frac{a}{x} - \frac{b}{1-x} - \frac{dc}{1+cx}\right) S, \qquad \mathbb{E}_g R = u + vM. \tag{9.6}$$

Here R is rational with poles only at $0, 1, -1/c, \infty$ and with gR integrable; u, v are algebraic and the reduction is chosen with vanishing endpoint term. The baseline proves uniqueness of u, v and the corresponding algebraic-primitive criterion.

Proposition 9.9. *All polynomial expectations in this DBKZ family generate exactly $\overline{\mathbb{Q}}(M)$. Algebraicity of a rational expression in them is decided by reducing it to a rational function of the indeterminate M: it is algebraic precisely when that rational function is constant. Every central moment of order $n \geq 2$, and every cumulant of order $n \geq 1$, is transcendental. An algebraic primitive of gR exists precisely when both coefficients u, v in (9.6) vanish.*

Proof. Each raw moment is affine in M, and the first moment is M. A nonconstant rational function evaluated at a transcendental number is transcendental. The coefficient of M^n in the nth central moment is $(-1)^{n-1}(n-1)$, and in the nth cumulant it is $(-1)^{n-1}(n-1)!$. These are nonzero in the stated ranges. The primitive assertion is the two-dimensional de Rham reduction in the baseline. It is stronger than the definite-integral criterion $v = 0$, which permits a nonzero algebraic expectation. □

Two effective transfers require no new arithmetic input. For rational $s > 0$, $r > 1/2$, define the generalized Brouncker constant by

$$Y(s,r) = 4r\frac{\Gamma(A+1/2)\Gamma(B_1+1/2)}{\Gamma(A)\Gamma(B_1)}, \qquad A = \frac{s+1}{4r}, \quad B_1 = \frac{s+2r-1}{4r}.$$

The baseline gives this explicit finite classifier: if $2A$ or $2B_1$ is integral the value is transcendental; otherwise choose even N with $NA, NB_1 \in \mathbb{Z}$. The value is algebraic exactly when, for every k coprime to N, the fractional parts $\{kA\}$ and $\{kB_1\}$ lie in opposite open halves of $(0,1)$. Thus equality and sign of a polynomial in Y with real algebraic coefficients are decidable. In the algebraic cases one uses exact algebraic arithmetic; in the transcendental cases a nonzero polynomial cannot vanish, so certified interval approximation eventually decides its sign. The same criterion decides the minimality condition $v_0/v_{-1} = s - Y(s,r)$ in

$$(2rn+1)v_n = 2sv_{n-1} + (2rn-1)v_{n-2}, \qquad r > 1/2,\ s > 0,$$

and the corresponding sign-reversed positivity boundary $w_0/w_{-1} = Y(s,r) - s$, with $w_{-1} > 0$; the zero sequence is handled separately. These are decisions for the classified rational parameter family, not for arbitrary continued fractions or recurrences.

For derivative transfers put

$$C_h(c) = \frac{\int_0^1 x^h(1-x)^h(1+cx)^{-h-1}\,dx}{\int_0^1 x^h(1-x)^h(1+cx)^{-h}\,dx}, \qquad h \in \mathbb{Q}, \quad h > -1.$$

We use the baseline diagonal theorem: $C_h(c)$ is transcendental for algebraic $c > 0$. Its Riccati equation is

$$cC_h' = (h+1)/(1+c) - C_h - hC_h^2. \tag{9.7}$$

Proposition 9.10 (Derivative and inverse transfers)**.** *If $h \neq 0$ and $c > 0$ is algebraic, then $1, C_h(c), C_h'(c), \ldots, C_h^{(N)}(c)$ are linearly independent over $\overline{\mathbb{Q}}$ for every N, while their generated field has transcendence degree one. If $C_h(c_*) = r \in \overline{\mathbb{Q}} \cap (0,1)$, then c_*, $C_h'(c_*)$, and dc_*/dr are transcendental.*

For $F_h(c) = {}_2F_1(1, h+1; 2h+2; -c)$, suppose $h > -1$ has reduced prime denominator at least 5. For each $n \geq 1$, $F_h^{(n)}(c)$ is transcendental at all real algebraic $c > -1$, $c \neq 0$, except for at most n explicitly computable algebraic points, at which it is algebraic. In particular, $F_{-4/5}(3)$ is transcendental but $F'_{-4/5}(3) = -1/20$.

Proof. Successive differentiation of (9.7) expresses $C_h^{(n)}$ as a polynomial in C_h of degree $n+1$, with leading coefficient $n!(-h/c)^n$. Distinct degrees imply linear independence. An algebraic c_* would contradict the diagonal theorem. Substitution $C_h(c_*) = r$ in (9.7) writes $C'_h(c_*)$ as a nonconstant rational function of c_* with algebraic coefficients. It is nonzero by strict monotonicity, so its reciprocal is transcendental as well. For F_h, use

$$c(c+1)F'_h + [(h+1)c + 2h + 1]F_h = 2h + 1.$$

Induction gives $F_h^{(n)} = A_n F_h + B_n$ with rational functions over $\mathbb{Q}$ and $A_n = P_n(c)/[c^n(c+1)^n]$, where $0 \neq P_n \in \mathbb{Q}[c]$ has degree at most n. Nonvanishing follows also from the homogeneous solution $c^{-2h-1}(1+c)^h$, whose nth derivative has nonzero leading asymptotic coefficient $(-1)^n(h+1)_n$. The baseline transcendence theorem for F_h gives exactly the stated exceptional set. Direct substitution gives the displayed exception. □

Remark 9.11. The exclusion $h = 0$ in the first assertion is necessary: then $cC'_0 = (1+c)^{-1} - C_0$, so all jets lie in the span of $1, C_0$. The inverse assertion does include $h = 0$.

Another contiguous family is

$${}_2F_1(3/5 + n, 6/5 + n; 12/5 + n; -1/5), \qquad n \geq 0. \tag{9.8}$$

Every value is transcendental, and these values together with 1 span a space of dimension exactly three over $\overline{\mathbb{Q}}$. Indeed, for $H(c) = {}_2F_1(3/5, 6/5; 12/5; -c)$ the baseline separation of beta and four-branch-point periods gives linear independence of $1, H(1/5), H'(1/5)$. The hypergeometric equation reduces all higher jets to this span. The differentiated Euler integrals are, up to a nonzero algebraic factor, integrals of strictly positive rational weights. In their two-dimensional reduction the period coefficients cannot both vanish: an exact term with vanishing endpoints would have zero integral. Thus no higher derivative is algebraic.

Proposition 9.12 (The denominator-33 field refinement)**.** *Let $\mathcal{A}_{33}$ consist of the nonidentity algebraic expressions $Q = \Gamma(a/33)\Gamma(b/33)/[\Gamma(c/33)\Gamma(d/33)]$, with $1 \leq a \leq b < 33$, $1 \leq c \leq d < 33$, and $a + b = c + d \neq 33$, identifying reciprocal expressions. There are thirty such expressions by the baseline classification. Each quotient Q generates*

$$K = \mathbb{Q}(\zeta_{33})^+(3^{1/11}), \qquad [K : \mathbb{Q}] = 110.$$

Of their squares, ten have degree 55 and twenty degree 110. The normal closure is $\mathbb{Q}(\zeta_{33}, 3^{1/11})$, of degree 220, with Galois group $\mathrm{AGL}(1, 11) \times C_2$.

Proof. Write $F = \mathbb{Q}(\zeta_{33})^+$ and $Q = 3^{m/11}c$, where $c \in F$ and $11 \nmid m$, using Table 1. Here

$$t_j = 2\sqrt{3}\sin(j\pi/33) = -(1 + 2\zeta_{33}^{11})((-\zeta_{33}^{17})^j - (-\zeta_{33}^{17})^{-j}) \in F.$$

Every row follows directly from reflection and triplication,

$$\Gamma(j/33)\Gamma((33-j)/33) = \frac{\pi}{\sin(j\pi/33)}, \qquad \frac{\Gamma(j/33)\Gamma((j+11)/33)\Gamma((j+22)/33)}{\Gamma(3j/33)} = 2\pi\, 3^{1/2 - j/11}.$$

The cyclotomic computation gives ten distinct conjugates of c in every row. All conjugates are real, so taking eleventh powers preserves distinctness. Hence $\mathbb{Q}(Q^{11}) = F \subseteq \mathbb{Q}(Q)$. Since $c \in F$,

the field $\mathbb{Q}(Q)$ contains $3^{m/11} = Q/c$; Bézout's identity and $11 \nmid m$ then recover $3^{1/11}$. Thus every Q generates $F(3^{1/11})$, of degree $10 \cdot 11 = 110$.

The twenty cyclotomic automorphisms applied to $(Q^{11})^2$ give ten distinct conjugates in twenty cases and five in the remaining ten cases, as recorded in Table 1. These counts can be checked by squaring the expressions in Table 1, raising them to the 11th power, and reducing their displayed cyclotomic expressions modulo $\Phi_{33}(X) = \Phi_{11}(X^3)/\Phi_{11}(X)$, with substitutions $X \mapsto X^u$, $(u, 33) = 1$. If $F_0 = \mathbb{Q}((Q^{11})^2)$, then $c^2 \in F_0$: its conjugates over the real field F_0 have equal 11th powers and therefore are equal. Hence $\mathbb{Q}(Q^2) = F_0(3^{1/11})$, whose degree is $11[F_0 : \mathbb{Q}]$.

The normal closure contains ζ_{11} and F, whose compositum is $\mathbb{Q}(\zeta_{33})$. Over that cyclotomic field $X^{11} - 3$ is irreducible, for the valuation of 3 at a prime above 3 is 2, coprime to 11. Thus the normal closure has degree $20 \cdot 11$. Its group is $C_{11} \rtimes ((\mathbb{Z}/33\mathbb{Z})^\times)$, with action through reduction modulo 11; the order-two kernel acts trivially, giving the asserted direct factor. □

Table 1: Exact certificates $Q = 3^{m/11}c$ and square degrees for all thirty denominator-33 expressions.

$(a, b; c, d)$	m	c	$[\mathbb{Q}(Q^2) : \mathbb{Q}]$
$(1, 12; 3, 10)$	-1	t_{10}	110
$(2, 13; 6, 9)$	-2	t_9	55
$(3, 14; 8, 9)$	-3	t_8	110
$(4, 15; 7, 12)$	-4	t_7	110
$(5, 16; 6, 15)$	-5	t_6	55
$(5, 18; 6, 17)$	-5	$\dfrac{t_6 t_{16}}{t_{15}}$	110
$(1, 23; 3, 21)$	-1	t_{12}	55
$(4, 21; 7, 18)$	-4	$\dfrac{t_7 t_{15}}{t_{12}}$	110
$(2, 24; 6, 20)$	-2	t_{13}	110
$(3, 24; 8, 19)$	-3	$\dfrac{t_8 t_{14}}{t_9}$	55
$(3, 25; 9, 19)$	-3	t_{14}	110
$(2, 27; 9, 20)$	-2	$\dfrac{t_9 t_{13}}{t_6}$	110
$(4, 26; 12, 18)$	-4	t_{15}	55
$(1, 30; 10, 21)$	-1	$\dfrac{t_{10} t_{12}}{t_3}$	110
$(5, 27; 15, 17)$	-5	t_{16}	110
$(6, 28; 16, 18)$	5	$\dfrac{t_{15}}{t_5 t_6}$	110
$(3, 32; 12, 23)$	1	$\dfrac{1}{t_1}$	110
$(7, 29; 15, 21)$	4	$\dfrac{t_{12}}{t_4 t_7}$	55
$(6, 31; 13, 24)$	2	$\dfrac{1}{t_2}$	110
$(8, 30; 14, 24)$	3	$\dfrac{t_9}{t_3 t_8}$	110
$(9, 30; 14, 25)$	3	$\dfrac{1}{t_3}$	55
$(9, 31; 13, 27)$	2	$\dfrac{t_6}{t_2 t_9}$	110
$(12, 29; 15, 26)$	4	$\dfrac{1}{t_4}$	110
$(10, 32; 12, 30)$	1	$\dfrac{t_3}{t_1 t_{10}}$	55
$(15, 28; 16, 27)$	5	$\dfrac{1}{t_5}$	110

$(a,b;c,d)$	m	c	$[\mathbb{Q}(Q^2):\mathbb{Q}]$
$(17,28;18,27)$	5	$\dfrac{t_{15}}{t_5 t_{16}}$	55
$(18,29;21,26)$	4	$\dfrac{t_{12}}{t_4 t_{15}}$	110
$(19,30;24,25)$	3	$\dfrac{t_9}{t_3 t_{14}}$	110
$(20,31;24,27)$	2	$\dfrac{t_6}{t_2 t_{13}}$	55
$(21,32;23,30)$	1	$\dfrac{t_3}{t_1 t_{12}}$	110

Finally let $L_1 = F_{1/3}(1)$ and $L_2 = F_{2/3}(1)$. Writing $g = \Gamma(1/3)$, the baseline identities are

$$L_1 = 5 - \frac{40\pi^2}{3\,2^{2/3} g^3}, \qquad L_2 = \frac{21}{2} - \frac{7\sqrt{3}\,g^3}{6\,2^{1/3}\pi}.$$

The algebraic independence of π and $\Gamma(1/3)$ therefore implies that each of (π, L_1) and (π, L_2) is algebraically independent. This is a transfer of the known Chudnovsky theorem; it gives no claim about $\Gamma(1/5)$ or Euler's constant.

9.4 Geometry, stochastic processes, and physical models

Proposition 9.13 (Closed algebraic trajectories)**.** *Let C be a smooth projective curve over the real algebraic numbers, γ a compact real component, and $f : C \dashrightarrow \mathbb{R}^N$ an algebraic immersion without poles on γ. The length of $f(\gamma)$ is transcendental. The same holds for length in a positive algebraic metric regular on the curve. A nonsingular compact real algebraic orbit of an algebraic vector field has transcendental period, provided the field is everywhere nonzero along that orbit.*

Proof. The quadratic differential $q = \sum_i (df_i)^2$ has an algebraic square-root cover, on which $\omega = \sqrt{q}$ is a meromorphic algebraic differential. Following the positive branch along the real orbit, and going twice around if necessary, represents a positive integer multiple of the length as a nonzero closed period. The nonzero-period theorem for meromorphic algebraic differentials, a consequence of the analytic subgroup theorem [2], makes it transcendental. An algebraic metric changes q but not this argument. For a vector field choose the rational time form η satisfying $\eta(V) = 1$ on the curve. It has no pole on the orbit, and its positive closed integral is the period. The same theorem applies. □

This includes algebraic knots; for example the trigonometric algebraic parametrization

$$((2+\cos 3t)\cos 2t, (2+\cos 3t)\sin 2t, \sin 3t)$$

has squared speed $4(2+\cos 3t)^2 + 9$ and transcendental length. For a smooth embedded algebraic plane oval of length L, its complete two-sided tube of algebraic radius ε below the reach has area $2\varepsilon L$, hence transcendental area. This planar formula does not automatically settle the three-dimensional tube expression $\pi\varepsilon^2 L$. Polynomial Hamiltonian flows on regular compact level curves at algebraic energies are included in the period statement. Polynomial lemniscate perimeters, and hence finite polynomial escape level perimeters, are developed more precisely in Section 1. In the Mandelbrot parameter setting the finite iterates $p_0(c) = 0$, $p_{n+1}(c) = p_n(c)^2 + c$ are polynomials; this concerns their algebraic level curves, not the limiting fractal boundary.

The following direct formulas give several other transfers of the beta input. For the Jacobi diffusion generator

$$\mathcal{A} = \tfrac{1}{2}x(1-x)\partial_x^2 + \tfrac{1}{2}[(1-a) - (2-a-b)x]\partial_x,$$

the probability of reaching 1 before 0 is $U(x)/B$. Both boundaries are accessible for $0 < a, b < 1$, and an algebraic interior starting point gives a transcendental probability. For one-dimensional steady conduction with $\kappa(x) = x^{1-a}(1-x)^{1-b}$ and boundary values $0, 1$, the solution is $u(x) = U(x)/B$, the flux magnitude is $1/B$, and the minimum Dirichlet energy $\int_0^1 \kappa u'^2$ is $1/B$.

Proposition 9.14 (Common-period resistance networks)**.** *In a finite connected graph, let the edge resistances be $R_e = c_e B$, where $c_e > 0$ are algebraic and $B = B(a, b)$ is as above. Every effective resistance and every nonzero relaxation time, with positive algebraic vertex capacities, is transcendental. For the associated continuous-time random walk every positive-integer-order hitting-time moment between distinct vertices is transcendental. Its hitting probabilities and stationary probabilities are algebraic.*

Proof. The conductance Laplacian is $B^{-1}L_0$ with algebraic L_0. Effective resistances are positive algebraic multiples of B; nonzero relaxation times have the same form. For hitting-time moments, the finite linear systems $Qm_n = -nm_{n-1}$, with zero target boundary value and $Q = B^{-1}Q_0$, give $m_n = B^n a_n$ with positive algebraic a_n. The equations for hitting and stationary probabilities cancel the common time scale. □

For completeness, the finite-window exponential input from [15] says that

$$J_d(c)/J_d(1) \text{ is transcendental}, \qquad J_d(c) = \int_0^c e^{-x^{2d}}\,dx, \qquad d \geq 1,\ c \in \overline{\mathbb{Q}} \cap (0,1). \tag{9.9}$$

It follows from the E-function endpoint separation theorem; a finite mixture of M algebraically specified translated and rescaled kernels has at most $5M^2$ candidate algebraic cuts at which the normalized cumulative probability can be algebraic, as detailed in the baseline. This is a bound on possible exceptions, not an assertion that all candidate points actually are exceptions. The same input also gives a precise heat-time transfer. For a fixed algebraic $0 < c < 1$, put

$$R_c(t) = \frac{\int_0^c e^{-x^2/(4t)}\,dx}{\int_0^1 e^{-x^2/(4t)}\,dx}, \qquad t > 0. \tag{9.10}$$

Every solution of $R_c(t) = q \in \overline{\mathbb{Q}} \cap (c, 1)$ is transcendental. Indeed, if t were algebraic the rescaling $x \mapsto x/(2\sqrt{t})$ and the same endpoint separation would exclude an algebraic ratio. A solution exists because $R_c(t)$ is continuous, tends to 1 as $t \downarrow 0$, and tends to c as $t \to \infty$.

Proposition 9.15 (Exact quantum probabilities)**.** *On $[0, 1]$ the operator*

$$H_d = -\partial_x^2 + d^2x^{4d-2} - d(2d-1)x^{2d-2}, \quad \psi'(0) = 0, \quad \psi'(1) + d\psi(1) = 0,$$

has ground state $\psi(x) = e^{-x^{2d}/2}/\sqrt{J_d(1)}$ of energy zero. Its probability of lying in $[0, c]$ is transcendental at every algebraic $c \in (0, 1)$. If this probability is p, then the two-mode state $\sqrt{p}\,|10\rangle + \sqrt{1-p}\,|01\rangle$ has transcendental reduced purity, linear entropy, and quantum Fisher information for a relative phase encoded on one mode. No finite circuit with algebraic initial amplitudes, algebraic gates, and algebraic measurement operators can exactly produce a specified transcendental measurement probability.

Proof. Factorization with $A = \partial_x + dx^{2d-1}$ gives $H_d = A^*A$ with the stated boundary conditions and $A\psi = 0$. The Born probability is (9.9). The three quantities are $p^2 + (1-p)^2$, $2p(1-p)$, and $4p(1-p)$, respectively, nonconstant polynomials in a transcendental p. A finite algebraic circuit retains algebraic amplitudes and hence algebraic probabilities for computational-basis, or more generally algebraic-operator, measurements; finite adaptivity and nonzero postselection preserve algebraicity by sums, products, and quotients. Approximation is not excluded. No von Neumann or Shannon entropy claim follows from the polynomial argument. □

9.5 Numbered catalogue of transfers 1–60

The identifiers below preserve the discussion's numbering. "AI" means algebraic independence over $\overline{\mathbb{Q}}$; adjoining e is included where specified. "T" means transcendental. Statements about beta tilts use (9.3); a reference to the baseline means that the arithmetic theorem is an input from [15], not newly proved by the transfer. The later geometric developments refine item 21.

No.	Audited statement and scope	Proof or input
1	For fixed order, all Gaussian nodes and weights of μ, together with e, are AI.	Proposition 9.2
2	Normalized Gaussian coordinates are AI after deleting one weight; the full list has only the sum-of-weights relation.	Proposition 9.2
3	All finite lists of positive Hankel determinants, together with e, are AI.	Proposition 9.2
4	The nonleading coefficients of one fixed-degree monic orthogonal polynomial, together with e, are AI.	Proposition 9.2
5	Finite Jacobi recurrence coordinates and the mass, together with e, are AI.	Proposition 9.2
6	The corresponding Stieltjes Padé poles and residues, together with e, are AI.	Proposition 9.2
7	Every Bernoulli-updated beta posterior in the stated family has T interior algebraic-probability quantiles.	Proposition 9.7
8	Nonnegative polynomial likelihoods preserve this quantile conclusion.	Proposition 9.7
9	Dirichlet aggregate margins and count updates have T quantiles when the resulting two parameters meet the stated beta hypotheses.	Section 9.2
10	At algebraic interior cuts, posterior cumulative, survival, and odds values are T.	Proposition 9.7
11	Full polynomial moments and finite polynomial predictive probabilities are algebraic; they are rational for rational polynomial weights.	Lemma 9.6
12	Order-statistic algebraic-level quantiles are T. Separately, the minimum of two standard Cauchy variables has a T median.	Section 9.2; see text below
13	Iid coherent reliability systems inherit T cumulative values at algebraic cuts and T interior algebraic-level quantiles.	Section 9.2
14	Density, hazard, and reverse hazard at algebraic cuts are T if the polynomial weight is nonzero there.	Proposition 9.7
15	Positive-order one-sided conditional moments and the corresponding mean residual lifetimes at algebraic cuts are T.	Proposition 9.7
16	Gaussian quadrature for the beta tilts has algebraic nodes and weights, but its atomic cumulative function cannot equal the exact cumulative value at an algebraic interior cut.	Section 9.2
17	Equal-mass grids and the indicated monotone transport values are T.	Section 9.2
18	Even-integer-β Jacobi ensembles have T expected cumulative eigenvalue fractions at algebraic interior cuts.	Theorem 9.8
19	Student quantiles: even integer degrees of freedom give algebraic quantiles at algebraic probabilities; odd degrees at least three give T nonmedian quantiles. For one degree, algebraicity occurs exactly at rational probabilities.	Baseline, Student classification
20	For beta shapes $\alpha, \beta > 1$ rational with integer sum, shortest fixed algebraic-mass intervals have algebraic endpoints and width when both shapes are integers; otherwise endpoints and width are T with joint transcendence degree one. In this latter case the midpoint is $1/2$ for equal shapes and T otherwise.	Baseline, HPD classification
21	Every polynomial lemniscate component at a positive algebraic level has T length.	Section 1
22	Finite polynomial escape-level and finite Mandelbrot-iterate level perimeters are T.	Section 9.4

No.	Audited statement and scope	Proof or input
23	Even-power oscillator periods and nonzero travel times between algebraic positions on a regular real branch are T.	See text below
24	Internal rational quarter-phase positions and velocities of the normalized even-power oscillator are algebraic for powers $2, 4, 6$, and T for even powers at least 8.	Theorem 5.2
25	The specified Jacobi diffusion has T two-boundary exit probabilities at algebraic starting points.	Section 9.4
26	The beta-conductivity problem has T flux and minimum energy.	Section 9.4
27	An M-component algebraic finite-window exponential mixture has at most $5M^2$ candidate algebraic cuts with algebraic cumulative values.	Baseline, endpoint separation
28	Finite-window heat fractions at algebraic target levels in $(c, 1)$ can occur only at T times.	Equation (9.10)
29	The explicit factorized Schrödinger ground state has T Born probabilities at algebraic interior cuts.	Proposition 9.15
30	DBKZ polynomial expectation fields equal $\overline{\mathbb{Q}}(M)$; algebraicity of rational expressions is decidable by symbolic reduction.	Proposition 9.9
31	In that family, all central moments of order at least two and all positive-order cumulants are T; their field has transcendence degree one.	Proposition 9.9
32	The algebraic-primitive test is $u = v = 0$, whereas algebraicity of the normalized definite integral requires only $v = 0$.	Equation (9.6)
33	Equality and sign of algebraic-coefficient polynomials in the classified Brouncker constants are decidable.	Section 9.3
34	The stated two-parameter recurrence's minimality and associated positivity boundary are decidable at algebraic initial data in its classified rational-parameter family.	Section 9.3
35	For $h \neq 0$, the diagonal C_h derivative jets together with 1 are linearly independent, but have transcendence degree one.	Proposition 9.10
36	Algebraic target values of C_h have T inverse parameters, derivatives there, and inverse sensitivities.	Proposition 9.10
37	The nth derivative of the stated Log1 family has at most n algebraic exceptions; $F'_{-4/5}(3) = -1/20$ is one.	Proposition 9.10
38	Every value in (9.8) is T; together with 1 their linear span has dimension three.	Equation (9.8)
39	The thirty classified denominator-33 quotients have degree 110; ten squares have degree 55, twenty have degree 110; the normal closure has degree 220.	Proposition 9.12
40	Each of $(\pi, F_{1/3}(1))$ and $(\pi, F_{2/3}(1))$ is AI.	Section 9.3
41	$\log \mathrm{Ein}(1)$ and logarithms of positive nonconstant rational functions of the moment sequence are T.	Theorem 9.1; (9.2)
42	Integer-order Rényi entropies of Gaussian quadrature weights are T for at least two nodes and order at least two.	Proposition 9.3
43	Gaussian polynomial error is zero through degree $2q - 1$ and T for every algebraic-coefficient polynomial of higher degree.	Proposition 9.3
44	At one fixed real algebraic evaluation point, all finite lists of Christoffel extremal energies, together with e, are AI.	Proposition 9.3
45	The Vandermonde-square ensemble partition functions are AI, and their logarithms are T.	Proposition 9.4
46	Every nonconstant real algebraic-coefficient symmetric polynomial observable in that ensemble has T expectation and variance.	Proposition 9.4
47	All finite lists of normalized cumulants, together with e, m_0, are AI.	Proposition 9.4
48	The moment-generating function and its logarithmic derivative satisfy no nonzero algebraic differential equation over $\overline{\mathbb{Q}}(e)(t)$.	Corollary 9.5
49	A finite Jacobi Hamiltonian with at least three sites has no exact projective boundary-state revival at positive time.	Corollary 9.5

No.	Audited statement and scope	Proof or input
50	Integer-power asymmetric losses have algebraic minimizers exactly for an even power and equal side weights.	Theorem 9.8
51	The integer-order Wasserstein distance to an algebraic interior point mass is algebraic for even order and T for odd order.	Theorem 9.8
52	One Lloyd update from algebraic centers makes both outer centers T; stationary codebooks obey the stated outer-pair restrictions.	Theorem 9.8
53	Nonnegative rational multivariate polynomial tilts preserve rational joint moments but give T marginal algebraic-cut cumulative probabilities.	Theorem 9.8
54	All expected nonnegative-integer partial power sums in even-integer-β Jacobi ensembles are T at algebraic interior cuts.	Theorem 9.8
55	Compact immersed closed algebraic space curves, including algebraic knots and positive algebraic metrics, have T lengths.	Proposition 9.13
56	Thin two-sided planar tubes around embedded algebraic ovals have T areas at positive algebraic radii below the reach.	Section 9.4
57	Regular compact algebraic orbits of nonvanishing algebraic vector fields have T periods; polynomial Hamiltonian ovals are included.	Proposition 9.13
58	Common-beta-factor networks have T effective resistances, nonzero relaxation times, and positive-integer-order intervertex hitting-time moments.	Proposition 9.14
59	The spatially split quantum state has T purity, linear entropy, and one-mode relative-phase quantum Fisher information.	Proposition 9.15
60	Finite circuits with algebraic initial states, gates, and measurement operators cannot exactly generate a specified T measurement probability.	Proposition 9.15

Here are two details used in the catalogue. For item 12, the median of the minimum of two standard Cauchy variables corresponds to the one-variable cumulative level $1 - 1/\sqrt{2}$. Its value is $\tan(\pi(1/2 - 1/\sqrt{2}))$, which is transcendental by Gelfond–Schneider: algebraicity of the tangent would make the corresponding $e^{2\pi i(1/2-1/\sqrt{2})}$ algebraic, contradicting the theorem for an algebraic irrational exponent of -1. For item 23, normalize the energy equation as $\dot{x}^2 = 1 - x^N$, with N a positive even integer. For $N \geq 4$, $dx/\sqrt{1 - x^N}$ is holomorphic on its smooth projective hyperelliptic curve, so its nonzero algebraic-endpoint integrals are transcendental by the Abelian integral theorem. For $N = 2$ the integrals are nonzero algebraic logarithms, with the same conclusion by the logarithm theorem. Period and energy rescalings by nonzero algebraic factors preserve transcendence. The rational-phase division issue is substantially stronger than this forward travel-time assertion and is treated separately in the lemniscate division sections.

Remark 9.16 (Scope of the transfers). The catalogue concerns exact finite constructions under explicit hypotheses. Transcendence is not closed under limits, multiplication of unrelated transcendental constants, or arbitrary normalization by another period. The results do not prove irrationality of Euler's constant, transcendence of $\Gamma(1/5)$, general algebraic independence of periods, or a general statement about Shannon entropy. Distinct applications in the catalogue may encode the same arithmetic obstruction rather than sixty independent new transcendence theorems.

A Register of the final division and independence results

The transfer catalogue in the preceding part records observations 1–60, and the geometric register after Corollary 3.3 records 61–69. The remaining numbered developments are matched below to their precise statements. The entries share methods and implications; they are not being counted as independent solutions of unrelated open problems.

No.	Content and scope	Location
70	Rational division: elliptic reduction or a finite genus bound; the open-arc bound includes both endpoints.	Theorem 4.4.
71	Four independent genus-two integrals at every algebraic interior cut; normalized values and algebraic quantiles.	Theorem 7.7, Corollary 7.8.
72	gr independent moments on $r \le g$ ordered algebraic intervals of an explicit genus-g curve.	Theorem 7.5.
73	All integer-petal lemniscates: half-leaf classification, coordinate transcendence, and exact indexed division counts.	Corollary 5.9.
74	An explicit infinite independent family on one genus-two curve; also jointly independent with the Cassini family.	Theorem 7.11, Corollary 8.1.
75	The closed-oval bound $2g-2$ is attained in every genus; the open-arc bound is attained in genus two.	Theorem 4.12, Example 4.14.
76	Complete rational-quantile classification of $\mathrm{Beta}(\alpha, 1/2)$ for positive rational α; equivalent sinusoidal-spiral classification.	Theorems 5.1–5.2.
77	All algebraic division ratios for $n = 4$ and odd primes $n \ge 5$, using complete torsion packets.	Theorem 6.1.
78	For general regular differentials, algebraic fractions yield either the exact elliptic alternative or finitely many algebraic points.	Theorem 4.10.

B Boundaries of the conclusions and reproducibility

The rational-quantile classification does not imply its universal algebraic-ratio analogue: the torsion-packet theorem covers the specified prime cases and $n = 4$, not every composite index. Finiteness does not provide an effective enumeration or a genus bound for all algebraic ratios. Sharpness in every genus is proved for general regular differentials, not polynomial lemniscates.

Normalized closed traversals and branch-switching boundary routes differ. The smooth-boundary condition for areas is essential, as the area-2 critical Bernoulli lemniscate shows. Elementary integrability concerns a differential, not an individual value.

The E-function applications assert algebraic independence with their specified inputs; the Abelian-integral families assert linear independence. Neither type passes automatically to limits. No unresolved individual Euler or Gamma constant is settled.

The standalone LaTeX source embeds every section, table, proof, and bibliography entry. No external graphics, source fragments, or bibliography database is needed. Compile with `pdflatex` twice, or until references stabilize. Infinite families are given by exact equations or recursions; finite numerical lists are illustrative.

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